

Acquisition, (Mis)use and Dissemination of Information: The Blessing of Cursedness and Transparency^{*}

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Abstract

We study information policies in strategic settings in which agents observe an aggregate statistic of others' actions, but can only partially extract fundamental information from it. To model imperfect inference, we adapt cursed equilibrium to a simple beauty contest game with information acquisition, and discipline the information acquisition choice using a dogmatic envelope assumption, whereby agents correctly anticipate their actions but mistakenly deem them optimal.

Cursed agents acquire and disseminate more information, thereby creating a positive externality that, close to rationality, outweighs the losses arising from misuse. Reporting aggregate statistics more accurately (that is, increasing transparency) crowds out private information, but is the only policy that unambiguously raises welfare. By contrast, policies targeting fundamental information, such as acquisition subsidies or more precise fundamental signals, can backfire because they make cursed agents substitute away from an information source they already underuse.

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1 Introduction

Many economic decisions are taken in environments with interdependent payoffs and uncertainty both about fundamental states and the actions of others. To guide such decisions, players rely on information that includes direct sources about the fundamental as well as signals about aggregate statistics of the actions of others. Such statistics (“aggregative signals”) arise naturally in many economic settings, ranging from the transaction price in a financial market (Grossman and Stiglitz, 1980; Diamond and Verrecchia, 1981; Kyle, 1985) and the level of activity and number of infections during a pandemic, to inflation statistics (Lucas, 1972; Morris and Shin, 2005).

Aggregative signals do not just contain information about the actions of others but also disseminate information about the fundamental. The amount of fundamental information conveyed by such statistics depends both on their precision around the true aggregate moment and on the informativeness of the moment itself. Precision around the true aggregate moment, for which we adopt the moniker of transparency, may be reduced by measurement error, intentional coarsening of information, or delays in reporting. The informativeness of the moment, in turn, depends on how much private information the players use and disseminate through their strategies and ultimately on how much information they possess to begin with. These intricate interactions are at the centre of lively debates both in policy circles and in the academic literature, especially surrounding policy instruments that directly target the very availability and transmission of information.

One key though hitherto neglected aspect of these interactions is that extracting fundamental information from an equilibrium statistic is no easy feat. In particular, it requires understanding how the actions of others reflect their private information. Ample evidence (e.g. on the winner’s curse and underinference in social learning) indicates that agents often fail to take this inferential step.¹ A natural implication is that agents will also underuse aggregative statistics and place an excessive weight on private information.² This observation is empirically supported in several contexts.

¹In the winner’s curse, agents fail to appreciate that they are more likely to win a common value auction when their private information leads them to overestimate the value of the prize. Implicitly, they miss the information contained in winning the auction. See Kagel and Levin (2002) for a review of the experimental evidence. Weizsäcker (2010) shows through a meta-analysis that private information is overused in social learning experiments. This overuse extends to settings with continuous actions and stems from a failure to learn from observed actions rather than from others’ signals directly (Angrisani et al., 2021). In financial market versions of the social learning setting, there is evidence of contrarianism, i.e. buying at low and selling at high prices against the rational interpretation of information, consistent with an underappreciation that prices aggregate others’ private information (Cipriani and Guarino, 2005; Drehmann et al., 2005). See also Eyster et al. (2019, Section IV) for evidence of cursedness in financial markets. Most strikingly, Conlon et al. (2026) document that individuals learn less based on the information of others, even when they can observe others’ signals.

²Some experiments, by contrast, find an excessive use of social information. Participants choose to acquire information about the actions of others even though they are uninformative about the state in

Inflation expectations strongly depend on noisy private signals (individual supermarket prices) even when precise aggregative information is provided (Cavallo et al., 2017); traders in an experimental financial market fail to incorporate information from prices (Ngangoué and Weizsäcker, 2021). This evidence suggests that a complete analysis of such environments should account both for endogenous information dissemination and for agents' limited grasp of the information contained in aggregative signals. How does the processing bias affect the efficiency of information dissemination? What is the impact of information policies when information is endogenously acquired, disseminated, and misperceived? Which type of information should a policymaker provide?

To address these questions, we study a simple beauty contest game with information acquisition and adapt cursed equilibrium (Eyster and Rabin, 2005) as a tractable model of incomplete inference from aggregative signals. The beauty contest and its generalizations are a stylized yet rich laboratory used to study dispersed information across a range of applications (Hellwig and Veldkamp, 2009; Vives, 2017; Angeletos and Lian, 2018). Cursed equilibrium provides a parsimonious solution concept capturing the range from rational to fully cursed agents who fail to take into account that the actions of other players are a result of their private information and hence consider the aggregate outcomes to be uninformative about fundamentals. It has been used successfully to account for overbidding in common value auctions and has also found applications to financial markets (Eyster et al., 2019).³

Evaluating the effects of information policies requires a model which includes all the relevant adjustment margins. Those margins encompass both how agents use the information they possess and how much information they acquire: Reporting aggregate statistics more accurately—in our terminology, increasing transparency—may backfire because it crowds out private information acquisition and thereby undermines the very source of information in the aggregate statistic itself (Grossman and Stiglitz, 1980; Morris and Shin, 2005). To capture such adjustments, our beauty contest game features an information acquisition stage wherein agents choose the precision of their private information at a cost (as in Colombo et al., 2014).

To even represent this decision, one has to describe how agents assess the value of information ex-ante. While this is conceptually straightforward for rational agents,

a social learning game (Goeree and Yariv, 2015) and follow uninformative social signals in a financial contagion game (Trevino, 2020).

³Belief-based explanations of the winner's curse, among them cursed equilibrium, have been questioned by experiments showing that the phenomenon persists when the required beliefs are hard to rationalise or inconsistent with the rules of the game (Ivanov et al., 2010) and against computerised opponents using simple action rules (Charness and Levin, 2009). We take such behaviour to be consistent with cursedness in a wide sense, as a model for a lack of updating from difficult information sources requiring conditioning and inversion, including an inability to engage in conditional or hypothetical thinking (Esponda and Vespa, 2014; Ngangoué and Weizsäcker, 2021).

it becomes tricky when dealing with behavioural agents. Specifically, how do agents perceive the impact of altering the same information environment that they misperceive when acting within it? To what extent are they aware of their misuse, and how does such awareness affect information acquisition? Our modelling challenge is to address these issues with behaviourally plausible assumptions that yield a tractable analysis of the effects of information policies (e.g., increasing transparency) in environments with cursed agents. We assume that, at the acquisition stage, agents correctly anticipate the equilibrium relationships and their expected welfare, but do not regard their later use of information as erroneous. They therefore choose precision as if only its direct effect—obtaining more precise information while holding the action rule fixed—mattered, applying the logic of the envelope theorem dogmatically in a setting where it does not apply because information is later used incorrectly. We refer to this as the Dogmatic Envelope Assumption.

With this notion at hand, we can turn to the analysis of the model. The equilibrium is characterised by a vector of loadings on the different sources of information. As agents become more cursed, they substitute away from the aggregative signal and increase the use and acquisition of private information. This is because cursedness makes them perceive the aggregative signal as less informative, so they need to rely more on their remaining information sources. As a result, cursedness increases the inflow of private information into the aggregative signal. At the same time, it hinders its extraction. With information acquisition, these forces balance exactly: Contrary to the intuition that inferential naivety hampers information aggregation (see e.g. the results with exogenous private information in [Eyster et al. \(2019\)](#)), the covariance between the aggregate action and the state is independent of cursedness and informational efficiency even increases.

The core of our investigation is analysing the welfare consequences of cursedness and of information policies. In the rational equilibrium, the sensitivity of individuals' actions to (i.e., their use of) private information and the precision they acquire is inefficiently low because of an information dissemination externality. If agents are cursed enough, however, they may use and acquire private information at or even above the efficient level, though they simultaneously misuse their signals. As a result, the equilibrium is always inefficient. Welfare is nonmonotonic in the degree of cursedness. Local to rationality, *cursedness is bliss*: An increase in cursedness causes a (first-order) welfare gain from improved dissemination that dominates the (second-order) welfare loss from privately suboptimal use. Eventually, cursedness reduces equilibrium welfare as agents are increasingly unable to reap the gain from increased dissemination and significantly misuse their information.

Lowering the information acquisition costs and providing more public fundamental information can backfire. Although both policies increase welfare in the rational bench-

mark for our payoff specification (Bayona, 2018; Colombo et al., 2014), they can reduce welfare with cursed agents: They cause agents to substitute away from the aggregative signal and towards fundamental information (private or public, resp.), exacerbating suboptimal information use. We show that this effect can dominate the direct effect of cheaper and more precise information, causing the paradoxical comparative statics. Providing more aggregative information—increasing transparency—by contrast, is the only information policy that increases welfare across the whole parameter space: *transparency is bliss* because it improves the dissemination of information and does not exacerbate cursed agents’ misuse. Somewhat paradoxically, therefore, only increasing the precision of the very source of information that cursed agents underestimate makes them systematically better off.

We conclude the introductory section by discussing the related literature. In Section 2 we present the model and establish existence and uniqueness of a cursed equilibrium, taking the precision of private information as given. We introduce information acquisition in Section 3. We analyse the positive comparative statics of the model in Section 4. We turn to welfare analysis in Section 5. We discuss alternative notions of information acquisition and how our insights transfer in Section 6. Section 7 concludes.

1.1 Related Literature

The impact of information policies, be they about the fundamentals of the economy or about endogenous statistics, is at the heart of several research areas of applied economics. Working within the workhorse class of linear-quadratic models, we connect to a rich theoretical and applied literature ranging from business cycles (e.g. Hellwig and Veldkamp, 2009; Angeletos and La’O, 2010; Angeletos and Lian, 2018; Baeriswyl et al., 2020) and demand function competition (e.g. Vives, 1988, 2017), to political economy (e.g. Shadmehr et al., 2022). Angeletos and Pavan (2007) and Ui and Yoshizawa (2015) characterise the inefficiencies of information use in a general linear-quadratic Gaussian game. Colombo et al. (2014) study how private information acquisition affects the value of information. This literature generally considers only signals of the fundamental. We analyse the value of information in the presence of a signal of the average action, providing information of endogenous precision about the state. Bayona (2018) considers an information structure with such a signal in a setting akin to Angeletos and Pavan (2007), establishing that this can lead to a dissemination inefficiency in the use of private information.⁴ We focus on the role of agents’ limited

⁴Our rational benchmark nests a payoff restriction of Colombo et al. (2014), Bayona (2018), and their so far unexplored meet and establishes that the dissemination inefficiency goes hand in hand with inefficiently low information acquisition. Amador and Weill (2012) show that with such a dissemination externality more public information can cause a decrease in welfare, even without interdependent payoffs. Hebert and La’O (2023) study a model with flexible information acquisition, studying which cost functions lead to efficiency and non-fundamental volatility, and show that such an externality arises

understanding of aggregative information for the social value of information and transparency. Our restriction of the payoff structure to the simple beauty contest game allows us to isolate the novel sources of inefficiency in our setting.

The results of [Morris and Shin \(2002\)](#), who show that more precise public information can reduce welfare, have spurred extensive debate in the literature about the desirability of public information in particular in the context of central bank announcements.⁵ This discussion has often been couched in the terminology of “anti-transparency” vs. “pro-transparency”. This label does not correspond to our usage, as we reserve the word transparency for the precision of the public signal about the aggregate action. Although we would argue that much of the information provided by central banks is aggregative in nature, we also contribute to the original debate by demonstrating a novel channel based on cursed inference which can render public *fundamental* information undesirable: It distracts behavioural agents from other information sources whose information content they underestimate. The issue of endogenous information dissemination has been studied in the context of business cycles by [Wong \(2008\)](#) who shows that increased transparency can be self-defeating as it reduces the information available to the central bank itself to learn about the state of the economy, a mechanism that has also been studied in [Morris and Shin \(2005\)](#) and [Baeriswyl et al. \(2020\)](#).

Learning from a signal that aggregates individual best responses is also at the centre of the literature on information aggregation in financial markets. [Grossman and Stiglitz \(1980\)](#) show that the equilibrium informativeness of the price system is unresponsive to changes in transparency: an increase in noise leads to more information acquisition which exactly offsets the direct effect. We establish a similar invariance along a metric of informational efficiency in our setting.⁶ Updating biases have received considerable attention in the literature on behavioural finance, see [Barberis \(2018\)](#) for a recent survey. While this literature focuses on the time series properties of prices and returns, we

whenever the ease of acquiring information through the aggregate action depends on its responsiveness to the state, as is the case in our setting with additive noise.

⁵[Svensson \(2006\)](#), e.g., argues that the ratio of private to public precision required for the paradoxical welfare result is unreasonably high and [Woodford \(2005\)](#) calls into question the assumptions on strategic complementarity and welfare. The role of these assumptions is clarified and general conditions for such welfare results are given in [Angeletos and Pavan \(2007\)](#) and [Ui and Yoshizawa \(2015\)](#). [Kool et al. \(2011\)](#) show that public information can reduce information acquisition by market participants and thereby increase financial market volatility. [Amador and Weill \(2010\)](#) show that through a signal jamming channel public information can be welfare decreasing, as it reduces the informativeness of the price system thereby increasing uncertainty about the monetary shock.

⁶This feedback also creates widespread problems for equilibrium existence; usual ways to ensure existence are noisy information aggregation or a private values component ([Grossman and Stiglitz \(1980\)](#), see also [Vives \(2011\)](#) for the case of supply function competition). [Bayona and Manzano \(2024\)](#) and [Cohen \(2026\)](#) show that cursedness ensures existence even without these features. Our model has noisy information aggregation, as the average action is observed with noise. In a technical Appendix (C.4), we do study the case without noise, and show that an equilibrium does exist in the limit economy if agents are sufficiently cursed, $\chi > \sqrt{c} \frac{\tau_\theta + \tau_y}{(1-r)}$.

focus on the (social) value of information.

Underinference from the actions of others has been documented across many settings in the lab and in the field, see [Eyster \(2019\)](#) for a survey of related evidence. Cursed equilibrium was proposed by [Eyster and Rabin \(2005\)](#) as a model of such underinference. [Cohen and Li \(2023\)](#) and [Fong et al. \(2023\)](#) extend cursed equilibrium to extensive form games. [Cohen \(2026\)](#) defines a dynamic cursed expectations equilibrium notion and applies it in a financial market setting. We discuss the relationship between our approach and these concepts in Section 6. Most closely related to this paper are [Eyster et al. \(2019\)](#) and [Bayona and Manzano \(2024\)](#). [Eyster et al. \(2019\)](#) apply a cursed analogue to rational expectations equilibrium in a trading game and show that cursed behaviour can explain excessive trade volume. [Bayona and Manzano \(2024\)](#) introduce cursed equilibrium in the context of supply function competition with unknown cost parameters. They show that cursedness (including complete cursedness) can increase efficiency by reducing market power. We adopt a stylised payoff structure to focus on the informational channels including information acquisition and on information policies, and show that an interior level of cursedness is best for welfare through its effect on information dissemination. In the discussion following Proposition 1, we connect their findings to ours.

2 The Model with Exogenous Private Information

The game has two stages: First, agents choose how much private information to acquire. Second, agents play a beauty contest game. We begin our description of the setting by presenting the game with *exogenous* precision of private information in this section. We then introduce information acquisition in Section 3.

2.1 Setting

Actions and Payoffs

There is a unit interval of agents indexed by $i \in [0, 1]$, playing a simple beauty contest game. Their payoff is given by

$$u(a_i, \bar{a}, \theta) = -(1 - r)(a_i - \theta)^2 - r(a_i - \bar{a})^2, \quad (1)$$

where $a_i \in \mathbb{R}$ is the action of player i , $\bar{a} = \int_i a_i di \in \mathbb{R}$ is the average action⁷ and $\theta \in \mathbb{R}$ is the state (or fundamental). We allow for both strategic complementarity ($r > 0$), and

⁷As is customary, we adopt a law of large numbers for the private signals, $\int_i s_i di = \theta$, as a convention, see [Vives \(2008, 10.3.1\)](#) for a discussion. One formal operationalisation of this is to view the integrals in the sense of Pettis, see [Uhlig \(1996\)](#).

substitutability ($r < 0$), and assume that complementarity is not too strong ($r < 1$) to ensure the existence of a unique interior linear equilibrium and a planner solution.

Information and Best Response

The state θ is drawn from the prior distribution $\mathcal{N}(0, \tau_\theta^{-1})$. Agents receive three conditionally independent signals: a *private fundamental* signal $s_i \sim \mathcal{N}(\theta, \tau_s^{-1})$, i.i.d. across agents with a precision τ_s that we will endogenise in Section 3; a *public fundamental* signal $y \sim \mathcal{N}(\theta, \tau_y^{-1})$; and a *public aggregative* signal $\hat{p} \sim \mathcal{N}(\theta, \widehat{\tau}_p^{-1})$, which stems from the observation of the aggregate action $\bar{a}$ and therefore has an *endogenous* precision $\widehat{\tau}_p$, which we will derive shortly.

The optimal action is given by

$$a_i(s_i, y, \hat{p}) = \arg \max_{a_i} \mathbb{E}_\chi [u(a_i, \bar{a}, \theta)] \quad (2)$$

where $\mathbb{E}_\chi$ is the χ -cursed expectation operator with respect to agent i 's information, which we will define shortly. As u is quadratic, (2) takes the linear best response form

$$a_i = (1 - r) \mathbb{E}_\chi(\theta) + r \mathbb{E}_\chi(\bar{a}) \quad (3)$$

and, as the state and the signals are jointly Gaussian, the expected state is linear in the signals as well. We therefore focus on linear equilibria and conjecture an optimal action rule of the form

$$a_i = \alpha_0 + \alpha_s s_i + \alpha_y y + \alpha_p \hat{p} \quad (4)$$

for some vector of loadings $\alpha = (\alpha_0, \alpha_s, \alpha_y, \alpha_p)$. Then, we can write the true aggregate action as

$$\bar{a} = \int_0^1 a_i \, di = \delta_0 + \delta_\theta \theta + \delta_y y + \delta_p \hat{p} \quad (5)$$

with aggregate weights $\delta = (\delta_0, \delta_\theta, \delta_y, \delta_p)$. In an equilibrium, therefore, consistency will require that $\alpha = \delta$. To save on notation, we discuss the action rule in terms of the equilibrium loadings δ , and identify the use of the signals with the magnitude of the respective loading.

Informativeness of the Aggregative Signal

We are now ready to derive the endogenous precision of the aggregative signal, $\widehat{\tau}_p$. The source of aggregative information is that the agents learn from a noisy signal of the average action, namely $p = \bar{a} + z_p$ where $z_p \sim \mathcal{N}(0, \tau_p^{-1})$ is independent of the state and the fundamental signals. τ_p denotes the exogenous precision of the aggregative signal as a signal of $\bar{a}$, capturing how well agents can observe others' average action, and is

our transparency parameter.

We can rewrite p as a signal about the state, conditional on δ , as follows

$$\hat{p} = \frac{1}{\delta_\theta + \delta_p} (p - \delta_0 - \delta_y y) \sim \mathcal{N}(\theta, \frac{1}{\delta_\theta^2 \tau_p})$$

since jointly normal variables remain jointly normal under linear transformations. Hence, under the condition that $\delta_\theta + \delta_p \neq 0$ (which we verify in equilibrium), p and $\hat{p}$ are informationally equivalent conditional on y and therefore $\widehat{\tau}_p = \delta_\theta^2 \tau_p$. The Bayesian posterior on θ can be written based on the three conditionally independent sources $(s, y, \hat{p})$ which determines the posterior on $\bar{a}$ through (5). The precision of the aggregative signal about the state, $\delta_\theta^2 \tau_p$, depends both on transparency τ_p and on the equilibrium loading δ_θ .

2.2 Cursed Equilibrium: Definition

As a model of the failure to update from observing the action of others, we adapt cursed equilibrium (Eyster and Rabin, 2005). In this solution concept, agents are characterised by a parameter χ , the degree of cursedness, that ranges from $\chi = 0$ for the rational case to $\chi = 1$ denoting fully cursed behaviour. In the rational case, the agent forms his posterior in line with the structure of the model as $\mathbb{E}_{\text{rat}}[\theta] := \mathbb{E}[\theta|s_i, y, p] = \frac{\tau_y y + \tau_s s_i + \delta_\theta^2 \tau_p \hat{p}}{\tau_\theta + \tau_y + \tau_s + \delta_\theta^2 \tau_p}$. A fully cursed agent fails to perceive any correlation between other agents' actions and their private information. Instead, he thinks that others play according to the marginal distribution of their actions conditional on his fundamental information. Consequently, he thinks that the aggregative signal p (and equivalently $\hat{p}$) is conditionally uninformative about θ . A fully cursed agent therefore forms expectations according to $\mathbb{E}_{\text{cur}}[\theta] := \mathbb{E}[\theta|s_i, y] = \frac{\tau_y y + \tau_s s_i}{\tau_\theta + \tau_y + \tau_s}$.

Partially cursed agents are characterised by an interior level of cursedness $\chi \in (0, 1)$. They form expectations as a convex combination of rational and fully cursed ones, namely

$$\mathbb{E}_\chi[\theta] := \chi \mathbb{E}_{\text{cur}}[\theta] + (1 - \chi) \mathbb{E}_{\text{rat}}[\theta] = \chi \frac{\tau_y y + \tau_s s_i}{\tau_\theta + \tau_y + \tau_s} + (1 - \chi) \frac{\tau_y y + \tau_s s_i + \delta_\theta^2 \tau_p \hat{p}}{\tau_\theta + \tau_y + \tau_s + \delta_\theta^2 \tau_p} \quad (6)$$

$$\mathbb{E}_\chi[\bar{a}] := \delta_0 + \delta_\theta \mathbb{E}_\chi[\theta] + \delta_y y + \delta_p \hat{p} \quad (7)$$

Note that even cursed agents do all the updating about the state θ and then turn it into a belief about $\bar{a}$ through the equilibrium condition (5): they have “equilibrium awareness” in this sense. Where they go wrong is in the under-appreciation of the correlation between others' actions and private information.

Having introduced the updating bias, we can now define our equilibrium notion;

following Eyster et al. (2019) we adapt cursed equilibrium in a fashion similar to a linear rational expectations equilibrium. As is customary in the literature, we restrict attention to symmetric linear equilibria. Consequently, our claims to existence and uniqueness of equilibria refer to this class throughout.

Definition 1. A vector of loadings (α, δ) constitutes a χ -cursed expectations equilibrium if α satisfies the best response condition (3)-(4) with expectations formed according to (6)-(7) given δ , and the aggregate action is consistent with individual actions, $\delta = \alpha$.

2.3 Discussion

Before turning to the analysis of the equilibrium, we discuss our model and relate it to the literature and the applications it is aimed to capture.

Payoffs

Our choice of payoff specification puts us within a rich and familiar framework in which agents want to match a combination of the state and the average action (Morris and Shin, 2002; Angeletos and Pavan, 2007). They have found pervasive application, ranging from price setting games in which agents choose a price to match a fundamental demand state and the choice of competitors (Colombo et al., 2014, Sec. 4), demand function competition (e.g. Vives, 1988, 2017), to political economy (e.g. Shadmehr et al., 2022). We keep the analysis abstract by not adhering to any specific application and focus on the simple beauty contest payoff specification (1).⁸ The restriction to a simple beauty contest allows us to isolate the inefficiencies generated by the features specific to our information environment: the dissemination externality of aggregative information, and cursed updating from that source. Indeed, in our simple beauty contest game both information use and acquisition are efficient for the rational benchmark without aggregative information (Angeletos and Pavan, 2007; Colombo et al., 2014). Our aim is to work out the implications of information acquisition and dissemination by biased agents in a setting that is as simple and transparent as possible yet as rich as necessary in its basic ingredients. In light of the necessary complication this already entails, we abstract away from a general payoff structure (Angeletos and Pavan, 2007) required to concretely embed all applications. For example, in the context of a macro price-setting game, monopolistically competitive firms choose prices (a_i) to match an uncertain demand fundamental (θ) and the general price level $(\bar{a})$, where complementarity pricing

⁸Notice that our payoff specification differs from that in Morris and Shin (2002), who study the beauty contest utility $u_i((a_j)_{j \in [0,1]}, \theta) = -(1-r)(a_i - \theta)^2 - r \int (a_i - a_j)^2 dj$, which results in a dependence of individual utility on the variance of others' actions. This is what gives rise to the possibility of the paradoxical effects of public information that are central to their analysis, but absent in our rational benchmark.

($r > 0$) reflects consumer demand substitution patterns. Our payoff specification is a reduction of the fully micro-founded payoff specification derived in (Colombo et al., 2014, Sec. 4), but it retains many of its key features.

Timing

At the core of our model is the endogenous creation and dissemination of information about the state. Agents' actions are informed by their private and public fundamental information, which is aggregated into the average action $\bar{a}$, which is observed with noise and itself informs actions, and so on. This information feedback is a crucial feature of the situations we aim to model; it also raises difficult modelling questions regarding the timing of the interaction.

In the model, agents observe a signal $\hat{p}$ about the fundamental with precision $\hat{\tau}_p$. From the perspective of each individual agent, this precision is taken as given. From the perspective of the economist and the planner, however, it is determined endogenously in equilibrium through $\hat{\tau}_p = \delta_\theta^2 \tau_p$, and this is the object that enters both the planner's problem and our positive and normative comparative statics. We therefore formulate equilibrium in the spirit of rational expectations equilibrium: agents optimise taking the aggregative signal as given, while in equilibrium that signal is generated by their collective behaviour. This equilibrium formulation is standard in environments with endogenous aggregate information, and can be interpreted by viewing the static interaction as the steady state of a dynamic environment in which the fundamental θ evolves slowly enough for agents to learn from the previous period's average action. Closest to our setting, Bayona (2018, see Section 3.4) adopts this formulation under a general payoff specification with rational agents and exogenous private-information precision, and Eyster et al. (2019) do so for cursed agents.⁹

We can also formulate the interaction as a Bayesian game in action schedules. In this formulation, after observing the fundamental signals s_i and y , each player chooses a schedule $\mathbf{a}_i$ mapping the endogenous signal p into an action $a_i \in \mathbb{R}$. Equilibrium requires these schedules to satisfy $\int \mathbf{a}_i(p) di + z_p \equiv p$. This formulation is equivalent to our equilibrium notion (see also Vives, 2017), and which one is more natural depends on the application. In financial markets, schedules can be interpreted as limit orders, and the 'crossing-out' of the aggregative signal p corresponds to a central market mechanism. In other environments—for example, in the price-setting game of Colombo et al. (2014, Sec. 4), or more generally when firms or workers respond to inflation—it is

⁹See Minelli and Polemarchakis (2003) for a general theory of rational expectations equilibria in games, and also Angeletos and Werning (2006), who follow this approach to analyse endogenous information in a global game. Their working paper version provides an alternative microfoundation with two groups of agents: early movers, who lack an endogenous signal, and late movers, who observe the average action of early movers, with the rational expectations equilibrium emerging as the share of early movers goes to zero.

more natural to think of agents as reacting to a realised aggregative signal. As those settings are closer in spirit to the beauty-contest applications we have in mind, this is the interpretation we adopt in presenting the model.

The Interpretation of Transparency τ_p

Transparency in our model is the precision with which agents observe the aggregate outcome generated by others' actions. One interpretation is a setting in which a central authority has exclusive access to the actions taken by market participants. The authority processes these data through statistical analysis, which is noisy because of missing data, imperfect reporting, and related frictions, and produces a report that is then observed by everyone: τ_p captures the accuracy of this process, making it a natural parameter for both positive comparative statics and policy analysis.

Reading the static model as a reduced-form representation of a dynamic environment, lower transparency can also capture delays and other imperfections in the way past actions are converted into current public information. In the price-setting example, for instance, τ_p captures the timeliness and precision of inflation statistics or aggregate price indices published by a statistical agency, which firms consult alongside their private demand information when setting prices. More broadly, τ_p measures how much access agents have to information about others' behaviour when they choose their actions, and therefore the extent to which they can condition their decisions on an aggregate outcome—for example, on the price when submitting orders in a financial market.¹⁰

2.4 Cursed Equilibrium: Characterisation

This section studies the equilibrium for fixed τ_s . An equilibrium is computed by matching coefficients in the best-response function (3).

Theorem 1. *There exists a unique χ -cursed equilibrium for any τ_s . The intercept is $\delta_0 = 0$, the loading on private information $\delta_\theta \in (0, 1)$ is the unique real solution to*

$$\delta_\theta = [1 - r + r\delta_\theta] \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s + \delta_\theta^2 \tau_p} \left(1 + \chi \frac{\delta_\theta^2 \tau_p}{\tau_\theta + \tau_y + \tau_s} \right), \quad (8)$$

and the loadings on the public sources of information are given by

$$\delta_y = \frac{\delta_\theta \tau_y}{(1 - r) \tau_s}, \quad \delta_p = 1 - \delta_\theta \frac{\tau_\theta + \tau_y + (1 - r) \tau_s}{(1 - r) \tau_s}. \quad (9)$$

¹⁰In that context, a fully transparent market would be one in which a trader knows the transaction price before submitting an order (or submits a limit order). A lower level of transparency would correspond to a market in which traders submit market orders observing the transaction price only noisily—for example, through the current price in a similar marketplace or through past prices.

The right-hand side of equation (8) has a natural interpretation: it denotes the optimal loading on private information given aggregate δ_θ . First, the private signal is valuable for predicting the state, with best-response weight $1-r$, as well as the aggregate action to the degree that it reflects the state (conditional on public signals), with best-response weight $r\delta_\theta$. Second, the relative precision of the private signal is the usual Bayesian weight $\frac{\tau_s}{\tau_\theta + \tau_y + \tau_s + \delta_\theta^2 \tau_p}$. Hence, the private signal is ignored ($\delta_\theta = 0$) only if it is pure noise ($\tau_s = 0$). This term also contains the information spillover effect: the more other agents use their private information (higher δ_θ), the more can be learned from the aggregative signal, which reduces the weight on the private signal. Third, the final term adjusts this weight as cursed agents fail to understand that the aggregative signal is informative about the state and therefore perceive the private signal to be *relatively* more informative. In the extreme case of $\chi = 1$, the agent ignores the aggregative signal and the two final factors simplify to $\frac{\tau_s}{\tau_\theta + \tau_y + \tau_s}$, the relative precision of the private signal as if there were no aggregative information. Therefore, transparency is without effect in the fully cursed equilibrium while, symmetrically, cursedness has an impact on the equilibrium only if there is an informative aggregative signal ($\tau_p > 0$). Cursedness manifests itself as a pure updating bias and only distorts inference from the aggregative signal. Absent such signal, cursed agents act just like a rational agent as they correctly interpret all fundamental sources of information.

At first sight, that cursedness matters only in the presence of an aggregative signal may be surprising when compared with the implications of cursed equilibrium in a common value auction. In the auction, there is no aggregative information available to the agent before he chooses his action but still cursedness impacts his choice. This, however, is a natural consequence of the payoff structure: In an auction, the agent considers his payoff conditional on winning the auction, which is exactly such an aggregative conditioning event. In our model, the payoffs themselves weigh all states equally ex-ante and there is no such “implicit conditioning” embedded in them.

The fully rational case was analysed in depth in a more general payoff specification in [Bayona \(2018\)](#). Let us briefly discuss the fully cursed case.

Corollary (Fully Cursed Equilibrium δ^{FC}). *The equilibrium with $\chi = 1$ has*

$$\delta_\theta^{\text{FC}} = \frac{(1-r)\tau_s}{\tau_\theta + \tau_y + (1-r)\tau_s}, \quad \delta_y^{\text{FC}} = \frac{\tau_y}{\tau_\theta + \tau_y + (1-r)\tau_s}, \quad \delta_p^{\text{FC}} = 0 \quad (10)$$

The role of strategic substitutability and complementarity is directly apparent in the fully cursed equilibrium. If there are no such strategic interactions, cursed agents weigh the two signals at their (mental) disposal according to their precision. Strategic complementarity shifts weight away from the private signal s_i and towards the public signal, y , while substitutability has the opposite effect.

The fact that the fully cursed equilibrium puts no weight on the aggregative signal deserves clarification. This does not follow from cursedness alone. Indeed, even for fully cursed agents, the aggregative signal, $\hat{p}$, remains a valid source of public noise, z_p , which can be used as a correlation device. Consequently, even fully cursed agents would incorporate $\hat{p}$ into their best response as long as others do so. However, as they can directly condition on the public signal instead which provides information about the state in addition to correlated noise, they always want to put a lower weight on $\hat{p}$ than others. Hence, we only obtain $\delta_p^{\text{FC}} = 0$ as the consequence of an unravelling argument set in motion by the interplay between equilibrium and cursedness.¹¹

3 Information Acquisition

In the first stage, agents simultaneously choose the precision of their private signal, τ_s , at cost $c\tau_s$. If our agents were Bayesian, the next steps of analysis would be straightforward, as they would maximise their expected welfare,

$$W = \mathbb{E} \left[-(1-r)(a_i - \theta)^2 - r(a_i - \bar{a})^2 \right] - c\tau_s.$$

The beliefs (and actions) of cursed agents, however, are specified ex-interim only. Hence, the expected welfare equation above is not fully specified. To study information acquisition in this setting, one must make additional structural assumptions: How do agents forecast their own action a_i and the average action $\bar{a}$? What is the ex-ante expectation operator $\mathbb{E}$? How do agents conceptualize the channels through which information acquisition affects expected welfare (i.e., how to take the derivative of welfare)? For none of these three questions does the literature offer an obvious, canonical answer. Because information acquisition is an essential adjustment channel for assessing the impact of information policies, which is the main applied focus of this paper, we must commit to specific assumptions on these three ingredients.

In this section, we first motivate our modelling choices and formally define cursed expectations equilibrium with information acquisition. We then characterise this equilibrium. After deriving its positive and normative implications in the following sections, we return in Section 6 to discuss the implications of alternative modelling approaches.

¹¹Vives (2017) considers limited inference, equivalent to fully cursed behaviour, in a Linear-Quadratic-Normal model of competition in supply schedules with unknown costs. Fully cursed traders in his setting do not ignore the noisy signal of fundamentals, the price, as it is directly payoff relevant.

3.1 The Value of Information for Cursed Decision Makers

First, we must specify how agents think about their future actions. We assume they correctly predict both their action rule and how much they will learn from the various signals they will receive. They also correctly predict the equilibrium mapping (5) and understand that their choice of signal precision does not affect the equilibrium loadings δ nor the precision of information acquired by others. In particular, they understand that they are small and that their contribution cancels out by a law of large numbers. Hence, the precision choice is a single-agent problem; agents don't need to engage in genuine strategic reasoning at this stage. This assumption is natural in the applications typically analysed within the beauty-contest framework, such as price setting, where decision makers are small and typically understand how their own decisions respond to information and that their individual actions have no material effect on the aggregate objects.

Second, we need to choose the ex-ante measure over the state and signals. There are two natural candidates: the true ex-ante measure, and the ex-ante measure that generates and rationalises the cursed behaviour ex-interim ('quasi-Bayesian').¹² The characterisation of the latter beliefs, fully developed in Appendix C.1, uncovers a significant conceptual shortcoming, driven by a fundamental feature of cursedness. Cursedness posits that agents update their beliefs about the actions of others based on information about the state but are unable to update their beliefs about the state based on information about the actions of others. This asymmetry is at the core of the behavioural intuition of the bias; it is also inconsistent with a unified (potentially misspecified) Bayesian probabilistic model driving updating and behaviour.¹³ In our setting, this manifests itself as follows: Agents consider the aggregative signal p informative about the state from an ex-ante perspective, but uninformative about the state from an ex-interim perspective conditional on y, s_i . To achieve this, the perceived correlation between the state and the public aggregative signal needs to depend on the *individually chosen* τ_s ! In other words, using interim welfare to guide information acquisition would imply that an agent thinks that his personal information acquisition affects the precision of public information obtained by all agents. In short, the quasi-Bayesian approach implausibly makes each agent believe they are large enough to affect the quality of public statistics.

These conceptual problems rule out a quasi-Bayesian approach to information acquisition by cursed agents, so we turn instead to objective ex-ante welfare. This

¹²For a general link between updating biases and misspecification, see [Bohren and Hauser \(2023\)](#).

¹³In the original [Eyster and Rabin \(2005\)](#), this does not pose a problem, as in a one-shot environment the bias can be defined by a transformation of ex-interim beliefs, which are sufficient for the analysis. Taking cursedness into a genuinely dynamic setting, this becomes a modelling challenge which different contributions have tackled in different ways, see the discussion in Section 6.1.

criterion is particularly suited in settings that are experienced repeatedly, as it corresponds to the average experienced payoffs. It is therefore particularly suited to settings such as household decision making, price setting or retail investing, where agents may react instinctively to signals but consciously choose how much attention to devote to different sources of information based on their experience. Formally, we define our criterion as the ex-ante expected welfare according to the objective distribution, given an action rule α , an equilibrium δ , with private precision τ_s ; given by

$$W(\alpha, \delta, \tau_s) = -\frac{\alpha_s^2}{\tau_s} - \frac{1}{\tau_\theta} \left((1-r)(\alpha_s + \alpha_y + \alpha_p - 1)^2 + r(\alpha_s + \alpha_y + \alpha_p - \delta_\theta - \delta_y - \delta_p)^2 \right) \\ - \frac{1}{\tau_y} \left((1-r)\alpha_y^2 + r(\alpha_y - \delta_y)^2 \right) - \frac{1}{\delta_\theta^2 \tau_p} \left((1-r)\alpha_p^2 + r(\alpha_p - \delta_p)^2 \right) - c\tau_s \quad (11)$$

Once the objective is specified, we are left with the last question, namely through which channels agents perceive their information acquisition as affecting welfare. Differentiating (11) with respect to τ_s gives

$$\frac{d}{d\tau_s} W = \frac{\partial}{\partial \tau_s} W + \nabla_\alpha W \cdot \nabla_{\tau_s} \alpha,$$

which reveals two channels through which τ_s impacts expected welfare. The first term, $\frac{\partial}{\partial \tau_s} W$, is the direct effect: holding behaviour α fixed, it captures the (net) marginal benefit from more precise information. The second term, $\nabla_\alpha W \cdot \nabla_{\tau_s} \alpha$, is the indirect effect: it captures the welfare consequences of the agent's re-optimizing behaviour in response to the change in information precision. In a rational model (e.g. [Colombo et al., 2014](#), as well as in our $\chi = 0$ benchmark), this indirect effect can be safely ignored: by the envelope theorem, $\nabla_\alpha W \cdot \nabla_{\tau_s} \alpha = 0$, because for each τ_s the choice of α maximises W , so that $\nabla_\alpha W = \mathbf{0}$ at the optimum. In our setting, instead, α is the result of cursed reasoning and does not maximise the objective welfare W . As a result, the envelope theorem does not apply and the partial and the total derivative of welfare (generally) differ.

Choosing one over the other as a model of information acquisition is a behavioural assumption. Choosing the total derivative $\frac{d}{d\tau_s} W$ means modelling fully sophisticated agents; they understand how their information acquisition affects the distribution of signals, the induced actions, and hence welfare, and use information acquisition precisely to “heal” the welfare losses generated by cursedness at the information use stage. We view this as an implausible assumption for our applications. The agents we want to model, by contrast, update from information in a cursed manner *and consider doing so to be correct*. Making the mistake and not considering it to be in error are two sides of the same coin. What this means for information acquisition is that agents do not consider the indirect channel through which information acquisition affects their

welfare.

Our key assumption to operationalise this aspect of cursedness is that agents treat the envelope equality $\frac{d}{d\tau_s} W = \frac{\partial}{\partial \tau_s} W$ as a primitive functional identity.

Definition 2. Information acquisition satisfies the **Dogmatic Envelope Assumption (DEA)** if τ_s solves

$$\frac{\partial}{\partial \tau_s} W(\alpha, \delta, \tau_s) = 0. \quad (12)$$

Note that the DEA holds in any Bayesian model of information acquisition, regardless of whether agents are correctly specified or not: the welfare gradient is perceived as zero whenever ex-interim beliefs are derived from ex-ante beliefs via Bayesian updating. In our case, cursed updating requires a wedge between the ex-ante and ex-interim beliefs, and we maintain the DEA as a behavioural assumption, representing the idea that agents treat the effect of information acquisition mediated by their use as negligible to the first order, as if the action rule was chosen optimally.

3.2 χ -Cursed Expectations Equilibrium with Information Acquisition

The principles for disciplining information acquisition developed in the previous section yield the following equilibrium notion.

Definition 3. A tuple (α, δ, τ_s) constitutes a χ -cursed expectations equilibrium with information acquisition if (α, δ) constitute a χ -cursed expectations equilibrium given τ_s and (α, δ, τ_s) satisfy the dogmatic envelope assumption (12).

Following (12) and setting the partial derivative of (11) to zero yields a tight relationship between the use of private information, α_s , and the acquired precision,

$$\tau_s = \frac{\alpha_s}{\sqrt{c}}. \quad (13)$$

According to this condition, the weight on private information in the best response, α_s , is a sufficient statistic for the marginal value of private information, independently of the level of cursedness. Cursedness only affects the calculus through α_s which in turn depends on the equilibrium loadings δ .

Taking account of endogenous information acquisition in the equilibrium condition (8), we arrive at the key equilibrium equation with information acquisition,

$$\delta_\theta = [1 - r + r\delta_\theta] \frac{\delta_\theta}{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y + \delta_\theta^2 \tau_p)} \left(1 + \chi \frac{\sqrt{c}\delta_\theta^2 \tau_p}{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y)} \right). \quad (14)$$

Contrary to the setting with exogenous τ_s which always has an interior solution, the existence of an interior equilibrium with information acquisition requires a parametric

restriction. This is because we need to ensure not only that agents want to use their private information, but that they are also willing to acquire it. More precisely, we require that the best-response weight on private information (RHS) exceeds δ_θ local to $\delta_\theta = 0$. This is the case if

$$\sqrt{c} < \frac{1-r}{\tau_\theta + \tau_y} \quad (15)$$

or, equivalently, if the costs of acquiring information are sufficiently small compared to the benefits of private information in the trivial candidate equilibrium. These benefits depend on the precision of prior and public information $\tau_\theta + \tau_y$ and the relative value of public versus private information, as summarised by $1-r$.¹⁴ If this condition is not met, we are stuck in a corner solution with zero information acquisition (and therefore use). The trivial solution $\delta_\theta = 0$ of (14) describes the equilibrium in that case.

Theorem 2. *There exists a unique χ -cursed equilibrium with information acquisition. The acquisition of private information is proportional to its use according to (13).*

If $\sqrt{c} < \frac{1-r}{\tau_y + \tau_\theta}$, the loading on private information $\delta_\theta \in (0, 1)$ is the unique interior real solution to (14) and the loadings on the public sources of information are given by

$$\delta_y = \sqrt{c} \frac{\tau_y}{1-r}, \quad \delta_p = 1 - \delta_\theta - \sqrt{c} \frac{\tau_\theta + \tau_y}{1-r}. \quad (16)$$

Otherwise, we have a corner equilibrium with $\delta_\theta = \tau_s = \delta_p = 0$ and $\delta_y = \frac{\tau_y}{\tau_\theta + \tau_y}$.

To rule out trivial cases, we will from now on assume that condition 15 is satisfied.

4 Positive Comparative Statics

In this Section, we study the comparative statics of the model.

4.1 The Impact of Cursedness and Transparency

We begin by considering the impact of cursedness and transparency on the equilibrium loadings and information acquisition.

Proposition 1. *The equilibrium loadings and information acquisition respond to cursedness χ as follows*

$$\frac{d\delta_\theta}{d\chi} \geq 0, \quad \frac{d\delta_y}{d\chi} = 0, \quad \frac{d\delta_p}{d\chi} \leq 0, \quad \frac{d\tau_s}{d\chi} \geq 0. \quad (17)$$

¹⁴If we instead assume convex costs with an Inada-type condition at zero, the analogue of (15) is always satisfied and we have an interior solution.

The comparative statics w.r.t. transparency τ_p have the opposite sign, namely

$$\frac{d\delta_\theta}{d\tau_p} \leq 0, \quad \frac{d\delta_y}{d\tau_p} = 0, \quad \frac{d\delta_p}{d\tau_p} \geq 0, \quad \frac{d\tau_s}{d\tau_p} \leq 0. \quad (18)$$

Moreover, the precision of the aggregative signal increases in transparency, i.e. $\frac{d}{d\tau_p} \delta_\theta^2 \tau_p \geq 0$. The above inequalities are strict if $\tau_p \neq 0$ or $\chi \neq 1$, respectively.

The comparative statics describe the substitution across information sources as the (perceived) informativeness of the aggregative signal varies. If agents become more cursed, they load more on private information and less on the aggregative signal. As cursedness makes agents underestimate the information content of the aggregative signal, cursed agents substitute away from this source of information and towards fundamental information. Information acquisition amplifies this effect: Indeed, as a consequence of the dogmatic envelope assumption, τ_s grows proportionately to its use δ_θ and this increase in the precision of private information causes agents to load even more on this source of information. The loading on the public fundamental signal, δ_y , is invariant in cursedness.¹⁵

Conversely, an increase in transparency has the opposite effect on the equilibrium loadings as cursedness, formalising the intuition that cursedness and transparency are complementary antagonists. In particular, higher transparency decreases the loading on private information: As the private information of others is disseminated more effectively, each agent relies less on his own. This leads to a crowding-out effect as agents use and acquire less private information, which however never dominates the direct effect. The precision of the aggregative signal about the fundamental is always increasing in transparency (see Figure 1).

4.2 Information Structure and Strategic Environment

We now study how the use and acquisition of private information (δ_θ and τ_s) vary with the information and strategic environment.

Proposition 2. *The loading on private information responds as follows*

$$\frac{d\delta_\theta}{dc} < 0, \quad \frac{d\delta_\theta}{d\tau_y} < 0, \quad \frac{d\delta_\theta}{dr} < 0 \quad (19)$$

¹⁵This may be somewhat surprising, as the public fundamental signal is a closer substitute to the aggregative signal in a sense, both providing a public noise component. However, the comparative statics hides that the loading on y increases *indirectly*: since $\hat{p} = \frac{1}{\delta_\theta + \delta_p} (p - \delta_0 - \delta_y y)$ the decrease in δ_p implies a greater indirect loading on y .

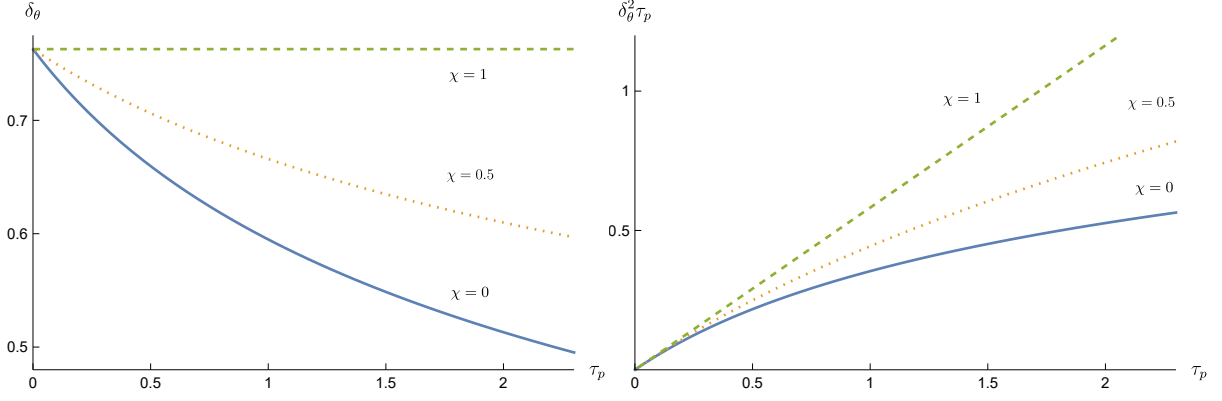


Figure 1: Reliance on Private Information vs. Crowd-Out: The effect of τ_p on δ_θ and $\delta_\theta^2 \tau_p$ for different levels of cursedness.

The comparative statics of τ_s have the same sign as the comparative statics of δ_θ , namely

$$\frac{d\tau_s}{dc} < 0, \quad \frac{d\tau_s}{d\tau_y} < 0, \quad \frac{d\tau_s}{dr} < 0. \quad (20)$$

As costs increase, agents acquire less precise private information and therefore rely on it less. When everybody does so, however, there is a counteracting force as the aggregative signal becomes less informative, leading to an increased reliance on private information as well as a boost in its value. This force never dominates and reliance on (as well as acquisition of) private information always decreases with cost. When the public fundamental signal becomes more precise, δ_θ decreases as agents shift away from the private signal. As complementarities become stronger the public signals y and p become more attractive relative to the private signal as they allow for better coordination with the aggregate action. Consequently, an increase in r decreases the weight on the private signal. Finally, the comparative statics on τ_s for all parameters but c follow directly from those of δ_θ since it is a sufficient statistic for the value of acquiring information by (13).

The loading δ_y on public fundamental information has intuitive comparative statics: agents load more on y if it is more precise, if private acquisition is more costly, and if the coordination motive is stronger. The loading on aggregative information instead has ambiguous comparative statics. Consider, for example, how δ_p responds to an increase in c . On the one hand, it reduces information acquisition and makes agents substitute towards the aggregative signal. On the other hand, the reduction in information acquisition and reliance on the private signal removes the very basis of information in p , making it less attractive. In Appendix A we show that either effect can dominate. A similar intuition is also the basis for ambiguous comparative statics of δ_p in the other parameters.

4.3 Overreaction vs. Underreaction to the State

We next turn to analysing how the aggregate action covaries with the state. Define β , the regression coefficient of the individual and aggregate action on the state,

$$\beta := \frac{\text{Cov}(\bar{a}, \theta)}{\mathbb{V}(\theta)} = \delta_\theta + \delta_y + \delta_p = 1 - \frac{\sqrt{c}\tau_\theta}{1-r}.$$

It is decreasing in c and r : cheaper fundamental information and stronger coordination incentives both lead to tighter comovement. Interestingly, however, it is invariant in both cursedness and transparency. These comparative statics of β reflect the exact balance of two forces. First, higher cursedness and lower transparency reduce the efficiency of information dissemination. For any level of informativeness of the aggregate action, agents extract less information from p and inject more noise. At the same time, they also increase the acquisition and use of private information, injecting more information into the aggregate action and the system of information dissemination as a whole. The invariance of β in τ_p echoes classic invariance results based on the crowd-out of private information acquisition ([Grossman and Stiglitz, 1980](#)), showing that they are valid even when agents underinfer from aggregate information. The invariance in the degree of cursedness implies that it cannot be identified based on the responsiveness of individual actions to fundamentals. Instead, it is identified by comparing the use of fundamental and aggregative signals.

Both invariance results stem from the combination of endogenous information acquisition and use. To isolate the information acquisition channel, we report the comparative statics of the model in which fundamental private information τ_s is exogenous (whose formal development is contained in [C.2](#)).

Claim 1. Suppose that τ_s is fixed exogenously. Then

- the comparative statics of [Proposition 1](#) still hold, except that $\frac{d\delta_y}{d\chi} \geq 0$ and $\frac{d\delta_p}{d\tau_p} \leq 0$;
- β becomes decreasing in cursedness and increasing in transparency,

strictly if $\tau_p \neq 0$ or $\chi \neq 1$, respectively.

The impact of cursedness on under- and overreaction has also been studied in related environments in [Eyster et al. \(2019\)](#) and [Bayona and Manzano \(2024\)](#). [Eyster et al. \(2019\)](#) show that cursedness reduces the responsiveness of the price with respect to the fundamental. This is in line with our prediction for the case without information acquisition; our results with information acquisition suggest that adding this channel may weaken this effect. [Bayona and Manzano \(2024\)](#) show that in their model of supply-function competition β may increase or decrease in cursedness. In their model, cursedness affects both learning from the price and, through a strategic effect, the extent of market power of the individual agents. Due to reduced market power,

supply functions become more responsive to private information, and this effect can dominate. With perfect competition, as in our atomistic setting, they show that there is no strategic effect and the responsiveness is decreasing in cursedness. We instead obtain a general invariance result through the information-acquisition channel, which is not present in either model. In our model, cursedness does not generate under- or overreaction; it changes the composition, but not the total amount, of information used by agents. In particular, more cursed agents substitute away from the indirect inference of disseminated private information (δ_p) and toward information they have acquired themselves (δ_θ).

4.4 Informational Efficiency

A central feature of our model is the aggregation and dissemination of privately acquired information, the efficiency of which is affected by cursedness. Following the literature in finance (Rostek and Weretka, 2012), we define the informational efficiency of the environment based on the reduction in uncertainty about the state based on observing the aggregative signal, $\hat{p}$, conditional on public information,

$$1 - \frac{\mathbb{V}[\theta|\hat{p}, y]}{\mathbb{V}[\theta|y]} = \frac{\delta_\theta^2 \tau_p}{\tau_\theta + \tau_y + \delta_\theta^2 \tau_p}.$$

Proposition 1 implies that informational efficiency increases in transparency. Even though there is a crowd-out effect ($\frac{\partial \delta_\theta}{\partial \tau_p} < 0$), it never fully offsets the mechanical increase of the precision of p . Informational efficiency is also increasing in cursedness. The increase in acquisition and use of private information as cursedness increases dominates the reduction in the efficiency of the recirculation of information.¹⁶

This finding contrasts with the literature on cursed traders. In Bayona and Manzano (2024), the analogous informational efficiency metric is constant in any symmetric linear equilibrium as there is a single source of noise, the traders' private signals. This result also carries over to the baseline specification in Eyster et al. (2019).

5 Welfare

Because informational efficiency is at most a partial guide to policy, we now turn to welfare. First, we characterise the first-best benchmark for the use and acquisition of information. Then, we use this benchmark to identify and characterise the inefficiencies

¹⁶Note that this informational efficiency metric doesn't fully capture the failure of cursed agents to extract this information. To address this margin, we may interpret the responsiveness coefficient β as an informational efficiency metric, as it measures how well the agents are able to align their action with the state.

of the χ -cursed equilibrium with information acquisition. This analysis lays the foundation for understanding the impact of cursedness and information policies on equilibrium welfare, which we turn to in the second part of this section.

In what follows, we let $W^{\text{EQ}} := W(\delta^\chi, \delta^\chi, \tau_s^\chi)$ denote equilibrium welfare, where we introduce δ^χ, τ_s^χ as shorthand for the equilibrium with χ -cursed agents.

5.1 The Planner Problem and the Inefficiencies of Equilibrium

As customary in the literature (see [Angeletos and Pavan, 2007](#)) our first-best welfare benchmark corresponds to the problem of a planner who controls both the use and acquisition of information, but cannot share information across agents. To this end we impose the consistency condition $\alpha = \delta$ in the welfare expression (11) and, with slight abuse of notation, let $W(\delta, \tau_s) := W(\delta, \delta, \tau_s)$ denote the objective of the planner, which can be expressed as

$$W(\delta, \tau_s) = -(1-r) \left\{ \frac{\delta_y^2}{\tau_y} + \frac{\delta_p^2}{\delta_\theta^2 \tau_p} + \frac{(1 - \delta_\theta - \delta_y - \delta_p)^2}{\tau_\theta} \right\} - \frac{\delta_\theta^2}{\tau_s} - c\tau_s. \quad (21)$$

Let the value of the planner problem be $W^* := \max_{\delta, \tau_s} W(\delta, \tau_s)$ and δ^*, τ_s^* the associated optimal loadings and precision. This optimum exists and is unique (Theorem 3 in the Appendix).

Two properties of the efficient solution follow readily from inspecting (21). First, the envelope theorem implies that the comparative statics of first-best welfare are all natural: As information is used efficiently by the planner, increasing precision—whatever the source—or lowering acquisition costs is always beneficial ($\frac{dW^*}{d\tau_\theta} > 0$, $\frac{dW^*}{d\tau_y} > 0$, $\frac{dW^*}{d\tau_p} > 0$, and $\frac{dW^*}{dc} < 0$). Second, maximising (21) with respect to τ_s gives the optimality condition $\tau_s^* = \frac{\delta_\theta^*}{\sqrt{c}}$, which coincides with condition (13): The use of private information is a sufficient statistic for the gains from acquiring it, also for the planner. Therefore, there is under-(over)acquisition of private information in equilibrium if and only if there is under- (over)use of private information in equilibrium. In particular, information acquisition in equilibrium is efficient if and only if information use in equilibrium is efficient.

Maximising (21) with respect to δ_θ shows that loading on private information in the efficient linear action rule is the unique solution of

$$\delta_\theta = (1-r+r\delta_\theta) \tau_s^* \frac{1}{\underbrace{\left(\frac{\tau_\theta + \tau_y + \tau_p \delta_\theta^2}{\tau_\theta + \tau_y + \tau_p \delta_\theta} \right)}_{\text{efficiency wedge}} + \tau_s^*},$$

where the efficiency wedge $\frac{\tau_\theta + \tau_y + \tau_p \delta_\theta^2}{\tau_\theta + \tau_y + \tau_p \delta_\theta} < 1$ represents how much the planner down-weights public information compared to the rational equilibrium (obtained by setting $\chi = 0$ in (8)). This implies that $\delta_\theta^* > \delta_\theta|_{\chi=0}$: the planner loads more on private information than the rational equilibrium to increase the dissemination of private information.¹⁷ Consequently, we find that the equilibrium is always inefficient: At $\chi = 0$, agents underuse private information as they do not account for the dissemination externality; at $\chi > 0$, they misuse their information sources. We illustrate these two sources of inefficiency in Figure 2, where we plot the gradient of welfare in equilibrium as a function of χ . In a fully rational equilibrium, the only externality is the dissemination of private information. Holding fixed the use of private information and thereby its dissemination, the other loadings of the equilibrium are conditionally efficient ($\frac{\partial W}{\partial \delta_y}(\delta^\chi, \tau_s^\chi) = \frac{\partial W}{\partial \delta_p}(\delta^\chi, \tau_s^\chi) = 0$ at $\chi = 0$). In a fully cursed equilibrium, agents ignore the aggregative signal altogether, so there is no dissemination externality and private information is used efficiently ($\frac{\partial W}{\partial \delta_\theta}(\delta^\chi, \tau_s^\chi) = \frac{\partial W}{\partial \delta_y}(\delta^\chi, \tau_s^\chi) = 0$ at $\chi = 1$).¹⁸ For any degrees of cursedness there is no externality or misunderstanding in the use of the public fundamental signal *relative* to the other signals.

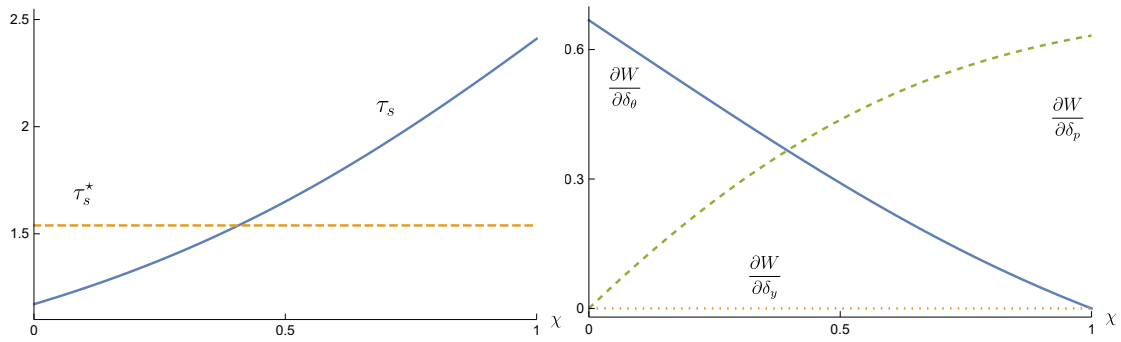


Figure 2: Equilibrium vs. efficient information acquisition (left) and the gradient of W in equilibrium (right) as a function of cursedness.

In the rational equilibrium there is also inefficiently little information acquisition, $\tau_s^* > \tau_s|_{\chi=0}$: rational agents disseminate too little private information both by under-using and by underacquiring it. Cursedness alleviates this inefficiency as it makes agents acquire more information (Proposition 1). Sufficiently cursed agents might acquire even above τ_s^* (see the left panel of Figure 2) if τ_p is large enough: Intuitively, information dissemination can be so effective that τ_s^* is low compared to the precision of information acquired in the fully cursed equilibrium.

Summarising the discussion above, we find that

¹⁷In a game of demand function competition, Vives (2017) shows that (rational) agents might place excessive weight on private information, because the information dissemination externality is counteracted by a pecuniary externality, which is absent in our setting.

¹⁸Recall that fully cursed equilibrium coincides at the action stage with fully rational equilibrium in which τ_p is set to zero. Without an aggregative signal our model is a special case of Angeletos and Pavan (2007) where payoffs satisfy the conditions for efficient use of information.

Proposition 3. *The equilibrium is always inefficient: for all primitives, $W^{\text{EQ}} < W^*$. Local to rationality there is inefficient use and acquisition of private information, but other sources are used efficiently. For positive levels of cursedness acquisition and private use can be at the efficient level, but the other sources are used inefficiently.*

5.2 The Comparative Statics of Equilibrium Welfare

The sources of equilibrium inefficiency identified above provide the bedrock for analysing the impact of cursedness and changes in the information environment on equilibrium welfare. Let $W^{\text{EQ}} := W(\delta^\chi, \tau_s^\chi)$ denote equilibrium welfare, where δ^χ, τ_s^χ is shorthand for the equilibrium with χ -cursed agents.

Cursedness is Bliss

We begin addressing the impact of cursedness on welfare. The value of the exercise is twofold. First, it helps us understand the impact of the behavioural bias on market outcomes and how it interacts with the inefficiencies of the rational benchmark. Second, as the literature has established that cursedness is codetermined by the environment, it can be in principle seen as a policy instrument.

This second point is most relevant in financial-market settings where our game is interpreted as one of submitting action schedules (see Section 2.3). Because the aggregate signal is unrealised when the schedule is chosen, considering its information content requires conditioning on a hypothetical future outcome. Experimental evidence (e.g. [Esponda and Vespa, 2014](#)) shows that such information is often only partly incorporated, and that the degree of rationality depends on the framing of the task and the market structure. In particular, [Ngangoué and Weizsäcker \(2021\)](#) show that traders extract less information from hypothetical prices (i.e., when placing a limit order) than from realised prices (i.e., trading at a given price). Our comparative statics therefore speak both to the desirability of policies addressing cursedness as an individual bias—for instance, through awareness or debiasing campaigns—and to broader interventions targeting features of the market structure that make inference from others' actions difficult.

Starting from the rational equilibrium, a marginal increase in cursedness affects welfare through two channels. First, agents now use their information suboptimally as they underestimate the information contained in p . The associated welfare reduction is second order, however, as δ is privately optimal in the rational equilibrium. Second, cursed agents acquire and disseminate more information. Since the rational equilibrium features underacquisition, this impact on the dissemination externality has a first order effect on welfare. Thus, local to rationality and up to first order, cursedness only has beneficial effects on welfare. When χ is already large, however, marginal increments in

cursedness have a first order negative effect from additional misuse, while the extent of underacquisition is lower. The effect of inefficient use dominates close to $\chi = 1$ as fully cursed agents do not use the information contained in the aggregative signal.

Proposition 4 (Cursedness is Bliss). *Equilibrium welfare is maximised for a unique interior level of cursedness: W^{EQ} is strictly quasi-concave in χ and*

$$\left. \frac{dW^{\text{EQ}}}{d\chi} \right|_{\chi=0} > 0, \quad \text{and} \quad \left. \frac{dW^{\text{EQ}}}{d\chi} \right|_{\chi=1} < 0. \quad (22)$$

Moreover $W^{\text{EQ}} \Big|_{\chi=0} > W^{\text{EQ}} \Big|_{\chi=1}$.

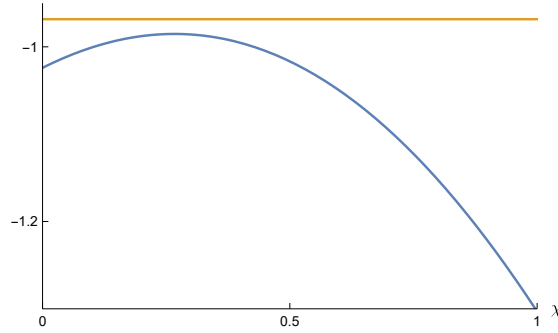


Figure 3: Equilibrium welfare as a function of cursedness χ (blue), compared to the efficient benchmark (yellow).

The shape of welfare as a function of χ and the comparison to efficient welfare is shown in Figure 3. As welfare is single-peaked in cursedness, the outcome of small policy experiments is informative about the position relative to optimal level. The fully rational equilibrium always outperforms the fully cursed case: Even though acquisition and dissemination of private information are higher, fully cursed agents are unable to make any use of their aggregative information. The inefficiently imprecise aggregative information provided in the rational equilibrium is preferable to complete ignorance of—albeit plentiful—aggregative information.

Transparency and Other Information Policies

Next, we turn to the impact of information policies. Given that cursed agents underuse aggregative information, what is the impact of providing them more precise information through different sources?

In the rational case, providing more information always increases welfare. This follows from our payoff specification and is an extension of the findings of [Colombo et al. \(2014\)](#) to a setting with endogenous information dissemination ([Bayona, 2018](#)). The same holds in the fully cursed case, except that a change in τ_p is without effect, as

aggregative information is always ignored. In general, however, information policies can backfire.

Proposition 5. *Equilibrium welfare W^{EQ} is always increasing in τ_p .*

W^{EQ} is increasing in τ_y, τ_θ and decreasing in c if χ is sufficiently close to either 0 or 1 or τ_p is sufficiently small. If, however, τ_p is sufficiently large, there exists an interior region of χ such that equilibrium welfare is

- (i) *decreasing in τ_y, τ_θ if strategic complementarities are sufficiently strong ($r > \frac{1}{2}$),*
- (ii) *and increasing in c in a game with strategic substitutes ($r < 0$).*

The proposition identifies sufficient conditions for counterintuitive comparative statics (exact conditions in the limit $\tau_p \rightarrow \infty$ are given in Lemma 4 in the Appendix). Let us first focus on the comparative static with respect to the precision of the public fundamental signal.¹⁹ An increase in τ_y has two equilibrium effects. First, a *direct effect*, as the precision of the agents' information increases mechanically. Second, a *substitution effect*, as agents reduce their use and acquisition of private information and also substitute towards y and away from p . When strategic complementarities are sufficiently strong, the second effect is particularly important and causes the paradoxical comparative static: For partially cursed agents, the use of p is already suboptimally low and a further decrease entails a welfare loss. This effect dominates the welfare calculus for interior χ . For (close to) fully cursed agents, however, the substitution effect is negligible as they disregard p and the direct effect is important as they rely heavily on y .

Likewise, an increase in acquisition costs potentially benefits partially cursed agents: It causes them to rely less on private information and to substitute towards p , whose informativeness they effectively underestimate. This effect can dominate with sufficiently strong strategic substitutes, when the value of information is relatively low because agents want to anti-coordinate. As we can see in Figure 4, the effect is present for sufficiently small costs, since then the substitution is towards a relatively informative p , whereas if costs are too high, the aggregative signal itself becomes too noisy.

A more precise aggregative signal (higher τ_p), by contrast, is always beneficial. The key observation to understand this difference is that despite the crowd-out of private information, transparency always renders the aggregative signal more informative

¹⁹A similar result is also obtained in Morris and Shin (2002), but for different reasons. There, all signals are fundamental, but the increased use of public information entails a payoff externality. In our setting, payoffs are such that—absent dissemination externality and cursedness—information use is efficient (Angeletos and Pavan, 2007; Colombo et al., 2014) and hence more precise public information is always welfare improving. Natural comparative statics continue to hold even if $\tau_p > 0$ and $\chi = 0$, a case not subsumed by the literature; therefore both the dissemination externality and cursedness interact to yield the paradoxical effects listed in Proposition 5.

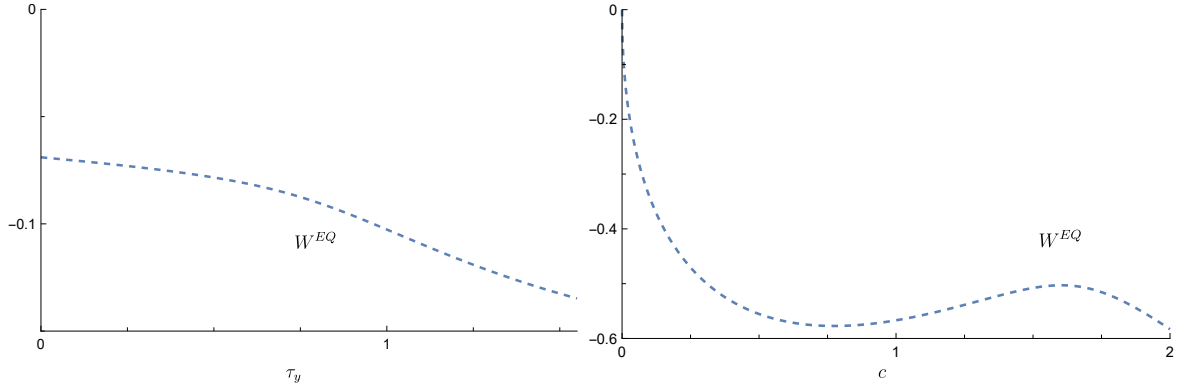


Figure 4: Counterintuitive comparative statics of welfare in τ_y (left) and c (right).

relative to the fundamental sources of information. Partially cursed agents therefore substitute towards the aggregative signal, which mitigates their bias. At worst, in the fully cursed case, the aggregative signal is not understood at all and hence irrelevant.

These findings can help to guide policy across our applications. For concreteness, let us focus on the implications in a macro price setting game akin to [Colombo et al. \(2014, Sec. 4\)](#). We find that even though agents underuse the information contained in aggregative sources, such as inflation reports from the central bank or statistical agency, it is always beneficial to provide more timely, precise, and fine-grained reports (which we interpret as increasing τ_p for the signal relevant to decision makers). This is the case even though agents fail to make a correct inference and extract all available information. While always beneficial, these policies may have a very small impact if agents are close to fully cursed, so their value needs to be judged against the costs of implementation on a case-by-case basis. Providing fundamental information about the state of the economy, for example through more precise consumer sentiment or foreign demand indicators, by contrast, can backfire. With public information, this danger is present when there are strong complementarities, arguably the relevant case in macro price setting games ([Morris and Shin, 2002](#); [Svensson, 2006](#)). In an environment where aggregative information is already abundant and precise (high τ_p), making fundamental public information more widely available may therefore generate harmful "cognitive crowd-out". Lowering the cost of acquiring private information about demand conditions can generate the same effect only when actions are strategic substitutes, so it is not a concern in this application.

However, since they misinterpret their information, cursed agents fail to reap the full benefits of transparency. This raises a natural question: might transparency raise redistributive concerns by favouring sophisticated market participants? In [Appendix B](#), we offer a preliminary examination of this issue by analysing the behaviour and welfare of an atomistic rational agent (shrewd agent) who can fully extract all available information but competes against a cursed crowd. The shrewd agent benefits from

the large amount of information disseminated by the cursed crowd, sometimes even abstaining from acquiring private information. However, he is also harmed by the crowd’s misuse of information, as strategic complementarity forces him to follow the crowd and distort his actions away from the fundamental. Compared to the rational environment, low levels of cursedness are always beneficial (indeed, the shrewd agent’s welfare can even exceed the first best), while for high levels of cursedness and strategic complements the need to imitate the cursed crowd can make the agent worse off. The trade-off between information free riding and miscoordination creates nontrivial comparative statics in the policy parameters: The shrewd agent always benefits from transparency but can be hurt by more public information and lower information acquisition costs—even when they are beneficial for the crowd.

6 Discussion of Alternative Approaches

In the main sections of the paper, we analyse our game using cursed equilibrium at the action stage and the dogmatic envelope assumption as the basis of the value of information. In this section we first discuss how recent notions of cursed equilibrium developed for dynamic games extend to our setting. We then study how our conclusions extend to alternative notions of the value of information considered in the literature, namely sophisticated agents and a representation as a misspecified Bayesian.

6.1 Sequential Cursed Equilibrium

Ultimately, we study the interaction between cursed players in a dynamic game. Two approaches have been proposed in the recent literature for a general extension of cursed equilibrium to dynamic games; neither is designed to analyse information acquisition by cursed agents. [Fong et al. \(2023\)](#) propose the notion of Cursed Sequential Equilibrium for multi-stage games with observed actions which therefore cannot be readily applied in our setting.

[Cohen and Li \(2023\)](#) propose sequential cursed equilibrium (SCE) as a solution concept for general extensive form games based on cursed equilibrium. To see what their notion implies in our setting, it is helpful to focus on two properties of SCE. First, players posit that the actions of their opponents are independent of each other (conditional on current information). Second, players do not expect to learn about the actions of others based on signals they themselves will receive in the future. What this implies in our setting is that the average action – from the perspective of the fully cursed agent – is deterministic at every stage. From an ex-ante perspective, the agent believes that the average action is deterministically zero; from the ex-interim perspective, he believes that it is deterministically at his interim expectation. Furthermore, the value

of information is zero as agents from an ex-ante perspective do not expect to learn based on their private information.

While these assumptions are well-founded for a portable model of cursedness that satisfies intuitive invariance properties across extensive forms, these implications seem implausible in our specific game and application. In a price-setting game, for example, it seems plausible to assume that managers expect to update their expectations about the price level when learning about local demand conditions, even though they fail to update about local demand conditions based on inflation statistics. This asymmetry is at the heart of our use of cursed equilibrium, but inconsistent with the symmetry between nature and other players in SCE. [Cohen \(2026\)](#) defines Dynamic Cursed Expectations Equilibrium in a similar spirit and analyses a financial market setting, establishing equilibrium existence when private information is sufficiently short-lived. Also in this notion, cursed players do not expect to learn from future signals, limiting the applicability when modelling information acquisition by cursed players.

6.2 Robustness to Alternative Notions of the Value of Information

To analyse the impact of underinference from aggregate statistics, we had to specify how cursed agents assess the value of private information. In this subsection, we study the two salient alternatives to our DEA notion and show, through numerical simulations, that our main policy insights extend, while the conceptual shortcomings of the alternative notions translate into distortions in information acquisition.

Sophisticated First, one could consider agents who choose τ_s to maximise true ex-ante welfare (11), that is, choose $\max_{\tau_s} W(\alpha(\tau_s, \delta), \delta, \tau_s)$, letting the action rule $\alpha(\tau_s, \delta)$ change in response to information acquisition. This corresponds to a rational agent who correctly predicts his cursed actions and desires to correct them by distorting the precision of private information available to his future biased self. As discussed in Section 3.1, this level of sophistication seems implausible to us in our applications; updating in a cursed fashion and not considering such an updating rule to be in error from an ex-ante perspective are two sides of the same coin.

Quasi-Bayesian Alternatively, we can consider our agents as misspecified Bayesians, their cursedness being the result of an incorrect prior belief about the joint distribution of states and signals ([Bohren and Hauser, 2023](#)). In particular, a χ -cursed agent corresponds to a Bayesian who believes that with probability $1 - \chi$ the signals will be drawn from the true distribution while with probability χ the signal will be drawn from a distribution that renders p uninformative about the state *conditional on* s_i and y (see Online Appendix C.1 for a formal development). This conditioning, inherent in the

notion of cursed equilibrium, is the reason the perceived correlation between the state and the public aggregative signal under this notion depends on the individually chosen precision of s_i . This approach is therefore inconsistent with agents understanding their (atomistic) position in the (large) economy, which is why in Section 3.1 we rejected this notion for our applications.

Despite these conceptual shortcomings, the two notions provide a useful benchmark and robustness exercise, as they span two focal ways to model ex-ante choice under cursedness: Agents may either recognise their future bias and respond to it strategically, or take their distorted interim beliefs including the implied effect of τ_s on the perceived informativeness of aggregate information as the correct object of optimisation.

Analysis First, note that for fully rational and fully cursed agents, both the sophisticated and the quasi-Bayesian solution coincide with our solution. In the rational case ($\chi = 0$), the action rule is correctly specified. Both the sophisticated and the quasi-Bayesian agents hence maximise objective welfare and so do agents following the dogmatic envelope assumption as it is justified as a consequence of the envelope theorem. More surprisingly, the same is true for fully cursed agents ($\chi = 1$). In that case, equilibrium behaviour is equivalent to that of a rational agent who is constrained not to use the aggregate signal, so conditional on this restriction the information-acquisition problem is again identical across the three notions. Hence, the choice between notions matters only for interior values of cursedness, where agents underreact to aggregate information but differ in how they conceptualise this distortion ex ante.

To illustrate the comparison with the dogmatic envelope assumption in the partially cursed case we resort to numerical simulation, as both the sophisticated and the quasi-Bayesian solution are less tractable and not the focus of our analysis. First, note that the equilibrium loadings in both alternative notions are close to the χ -cursed expectations equilibrium (Figure 5 right). The use of private information increases in all notions to counteract the reduction in the use of aggregative information. The key policy comparative statics results continue to hold. Cursedness is still bliss: Starting at $\chi = 0$, an increase in cursedness increases welfare (Figure 6 left). Moreover, for intermediate cursedness and high levels of transparency, more and cheaper fundamental information backfires (Figure 6). Higher transparency, instead, increased welfare in all our simulations.

The pattern of information acquisition depicted in Figure 5 also highlights why we set aside the sophisticated and quasi-Bayesian notions. In the sophisticated case, τ_s initially falls with cursedness: anticipating that his future self will over-rely on the private signal, the agent reduces its precision, thereby limiting misuse and forcing his future self to rely more on aggregative information (Figure 5, dotted line). Under the quasi-Bayesian notion, instead, agents acquire systematically more private information

than in our DEA benchmark (Figure 5, dashed line): this is because acquiring more private information increases the perceived precision of the aggregate statistic. As we approach full cursedness, agents cease to use the aggregative signal thereby neutralising this channel, which causes acquisition to fall locally as $\chi \rightarrow 1$ (the downturn of the dashed line). In both notions, we thus find that information acquisition is affected by channels we deem implausible in the context of our applications and these channels materially shape the equilibrium and the comparative statics, preventing a clean interpretation of the effects of information policies.

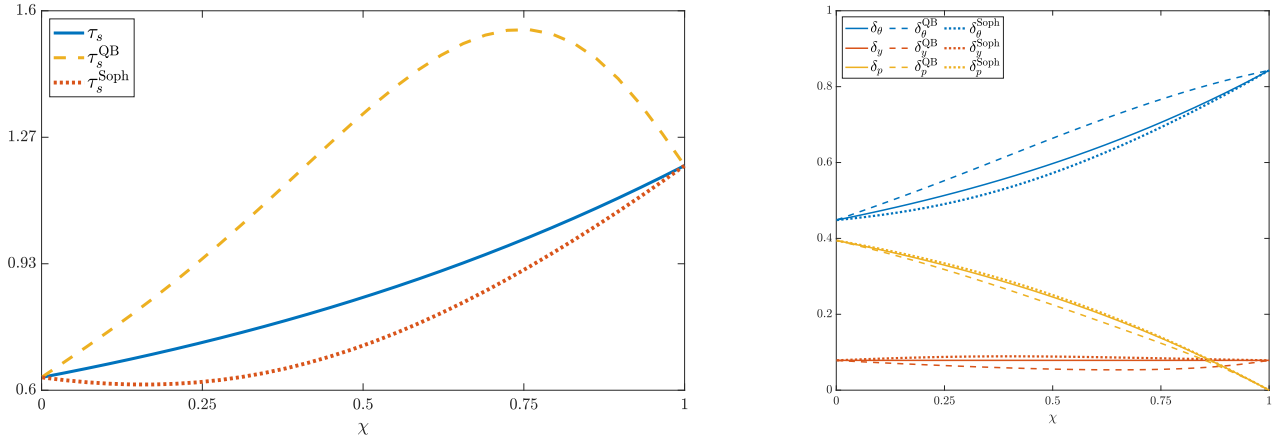


Figure 5: Dogmatic envelope (solid), quasi-Bayesian (dashed) and sophisticated (dotted): τ_s as a function of χ (left) and equilibrium loadings δ as a function of χ (right).

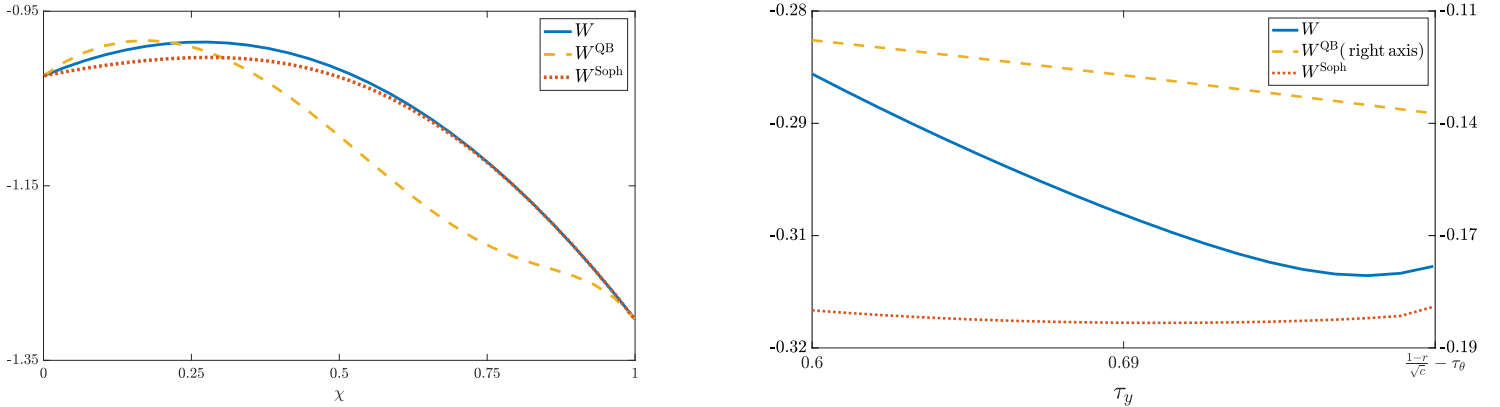


Figure 6: Welfare (dogmatic envelope (solid), quasi-Bayesian (dashed) and sophisticated (dotted)) as a function of χ (left) and as a function of τ_y (right).

7 Conclusion

We study the role of information policies when agents underinfer from aggregate statistics. We adapt cursed equilibrium to model underinference and introduce the

dogmatic envelope assumption to discipline information acquisition with misuse. The concept is based on the principle that agents correctly anticipate both the equilibrium strategies and how they will use their information, but mistakenly consider this use to be optimal.

In the rational benchmark there is inefficiently low acquisition and use of private information due to an information dissemination externality. Cursed agents rely more heavily on their private information, which can push information acquisition towards (or even above) its efficient level. While cursedness creates inefficiencies in information use, this increased dissemination of private information initially dominates: a bit of individual cursedness is a collective blessing. Providing more fundamental information can backfire as it makes cursed agents substitute away from an information source they already underuse. Increasing transparency, i.e. making the aggregate statistics that cursed agents underestimate more precise, by contrast, always improves welfare. Even though it crowds out the acquisition and use of private information, it always increases the endogenous precision of the aggregative signal. As a result it pulls cursed agents toward the information source they systematically underweight, mitigating the welfare loss from their bias.

However, even increasing transparency might have perverse redistributive effects in a world where some agents correctly interpret their information environment while interacting with a “cursed crowd”, as our extension in Appendix B indicates. Such shrewd agents benefit from transparency because they can free-ride on the crowd’s information, sometimes foregoing private acquisition altogether. While a full account of cognitive heterogeneity remains an important avenue for future research, these results, combined with our main analysis, already suggest that a policymaker considering information policies must weigh the risk of doing harm against the risk of having only modest effects which primarily benefit sophisticated players.

In sum, this paper shows that underinference from aggregative statistics is a key factor in determining the impact of information policies. We deliver these insights in a payoff specification in which all policies would have natural comparative statics without this ingredient (Colombo et al., 2014; Bayona, 2018). Our framework was kept minimal to deliver this insight in its most transparent form. Extending the analysis to more general payoff specifications as in Angeletos and Pavan (2007) and more deeply microfounded models yielding reduced forms similar to this class, as e.g. the business cycle model considered in Colombo et al. (2014) and demand function competition in Vives (2017) is a natural next step.

Appendix

We structure the Appendix as follows:

Appendix [A](#) provides the proofs of the results in the main text.

Appendix [B](#) studies the behaviour and welfare of an atomistic rational agent in the cursed economy.

We supplement this submission with an online appendix that develops extensions of the model referenced in the text and includes supporting proofs and algebraic computations referenced in the main appendix.

A Proofs

Throughout the appendix we use $\propto$ to denote equality of sign in equilibrium to drop positive factors in lengthy expressions. We use subscripts, e.g. f_{τ_y} , to denote partial derivatives.

We relegate lengthy computational proofs of auxiliary Lemmas used in this Appendix to Online Appendix [C.4](#).

Proof of Theorem 1: Recall from the text that $\delta_\theta = \alpha_s$ and the best response

$$a_i = (1-r) \left(\chi \frac{\tau_s s_i + \tau_y y}{\tau_\theta + \tau_y + \tau_s} + (1-\chi) \frac{\tau_s s_i + \tau_y y + \delta_\theta^2 \tau_p \frac{p-\delta_\theta-\delta_y y}{\delta_\theta+\delta_p}}{\tau_\theta + \tau_y + \tau_s + \delta_\theta^2 \tau_p} \right) + r \left(\delta_\theta + \delta_s \left(\chi \frac{\tau_s s_i + \tau_y y}{\tau_\theta + \tau_y + \tau_s} + (1-\chi) \frac{\tau_s s_i + \tau_y y + \delta_\theta^2 \tau_p \frac{p-\delta_\theta-\delta_y y}{\delta_\theta+\delta_p}}{\tau_\theta + \tau_y + \tau_s + \delta_\theta^2 \tau_p} \right) + \delta_y y + \delta_p p \right)$$

Matching coefficients with [\(4\)](#) and simplifying, we arrive at the desired expressions. From [\(8\)](#) we arrive at the equilibrium condition

$$0 = f(\delta_\theta) := \delta_\theta (\tau_\theta + \tau_y + \delta_\theta^2 \tau_p) - \chi \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} [1 - r + r\delta_\theta] \delta_\theta^2 \tau_p - (1-r)(1-\delta_\theta) \tau_s \quad (23)$$

To show that the solution is unique, note that $\frac{f(\delta_\theta)}{\tau_\theta + \tau_y + \tau_s + \delta_\theta^2 \tau_p}$ evaluated at $\delta_\theta = 0$ is equal to $-\frac{1-r}{\tau_\theta + \tau_y + \tau_s} < 0$ and at $\delta_\theta = 1$ it is greater than $1 - \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} > 0$. Hence, there is at least one root in $(0, 1)$. Furthermore, the expression is increasing at a root, as

$$\frac{\partial}{\partial \delta_\theta} \frac{f(\delta_\theta)}{\tau_\theta + \tau_y + \tau_s + \delta_\theta^2 \tau_p} = 1 - r \tau_s \left[\frac{\chi}{\tau_\theta + \tau_y + \tau_s} + \frac{1-\chi}{\tau_\theta + \tau_y + \tau_s + \delta_\theta^2 \tau_p} \right] + 2(1-\chi) \delta_\theta \tau_p \tau_s \frac{1-r+r\delta_\theta}{(\tau_\theta + \tau_y + \tau_s + \delta_\theta^2 \tau_p)^2}$$

where the first two terms are in sum positive and, at a root, we have $\text{sgn}\{\delta_\theta[(1-r) + r\delta_\theta]\} = 1$ whence the final term is also positive.

To transform from p to $\hat{p}$, we needed to presume that $\delta_\theta + \delta_p \neq 0$. To establish this, we will prove the stronger statement that $\delta_p \geq 0$. To see this, note that using (24)

$$\delta_p = 1 - \delta_\theta \frac{\tau_\theta + \tau_y + (1-r)\tau_s}{(1-r)\tau_s} = \frac{\delta_\theta(\tau_\theta + \tau_y + \tau_s)(\tau_\theta + \tau_y + \tau_s + \delta_\theta^2\tau_p)}{\tau_\theta + \tau_y + \tau_s + \chi\delta_\theta^2\tau_p} - \delta_\theta(\tau_\theta + \tau_y + \tau_s) \geq 0$$

□

Corollary 1. *In equilibrium, we have as an immediate consequence of $\delta_\theta > 0$ and (8) that*

$$1 - r + r\delta_\theta = \frac{\delta_\theta}{\tau_s \frac{\tau_\theta + \tau_y + \tau_s + \chi\delta_\theta^2\tau_p}{(\tau_\theta + \tau_y + \tau_s)(\tau_\theta + \tau_y + \tau_s + \delta_\theta^2\tau_p)}} > 0. \quad (24)$$

We now derive expected welfare, given in the text.

$$W(\alpha, \delta) = \mathbb{E}[-(1-r)(a_i - \theta)^2 - r(a_i - \bar{a})^2] = \mathbb{E}\left[-(1-r)(\alpha_s s_i + \alpha_y y + \alpha_p \hat{p} - \theta)^2 - r(\alpha_s s_i + \alpha_y y + \alpha_p \hat{p} - \bar{a})^2\right] \quad (25)$$

$$\begin{aligned} &= -\frac{\alpha_s^2}{\tau_s} - \frac{(1-r)(\alpha_s + \alpha_y + \alpha_p - 1)^2 + r(\alpha_s + \alpha_y + \alpha_p - \delta_\theta - \delta_y - \delta_p)^2}{\tau_\theta} - \frac{(1-r)\alpha_y^2 + r(\alpha_y - \delta_y)^2}{\tau_y} \\ &\quad - \frac{(1-r)\alpha_p^2 + r(\alpha_p - \delta_p)^2}{\delta_\theta^2\tau_p} \end{aligned} \quad (26)$$

This yields equation (11). Imposing $\alpha = \delta$, we get equation (21).

Proof of Theorem 2: Equation (13) is derived in the text assuming that $\delta_\theta \geq 0$. First, we need to show that there cannot be an equilibrium with $\delta_\theta < 0$, which is relegated to the appendix as Lemma 6.

Our equilibrium system is defined by

$$f(\delta_\theta, \tau_s) = \delta_\theta(\tau_\theta + \tau_y + \delta_\theta^2\tau_p) - \chi \frac{\tau_s[(1-r) + r\delta_\theta]\delta_\theta^2\tau_p}{\tau_\theta + \tau_y + \tau_s} - (1-r)(1-\delta_\theta)\tau_s = 0 \quad (27)$$

$$g(\delta_\theta, \tau_s) = \delta_\theta^2 - c \cdot \tau_s^2 = 0 \quad (28)$$

substituting g into f we obtain that loading δ_θ must solve

$$0 = \tilde{f}(\delta_\theta) := \delta_\theta + \sqrt{c}(\tau_\theta + \tau_y + \delta_\theta^2\tau_p) - [(1-r) + r\delta_\theta] \left(1 + \chi \frac{\sqrt{c}\delta_\theta^2\tau_p}{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y)} \right) \quad (29)$$

Lemma 1. *We have $\tilde{f}_{\delta_\theta} > 0$ for all $\delta_\theta > 0$.*

Note that $\tilde{f}(1) > \sqrt{c}\tau_p(1 - \frac{1}{1+\sqrt{c}(\tau_\theta+\tau_y)}) > 0$. Moreover, $\tilde{f}(0) = \sqrt{c}(\tau_\theta + \tau_y) - (1-r)$, which is negative if and only if condition (15) holds. In that case, therefore, a solution to (29) exists. Else, we are in a corner case $\delta_\theta = \tau_s = 0$. We obtain $\delta_y = \frac{\tau_y}{\tau_\theta + \tau_y}$ and $\delta_p = 0$ by plugging $\delta_\theta = \tau_s = 0$ into the original matching coefficients equations. Equation (16) follows immediately from plugging (13) into (9). $\square$

Proof of Proposition 1: Implicitly differentiating the equilibrium system (27)-(28) we get, for a generic parameter v

$$\frac{d\delta_\theta}{dv} = \frac{g_{\tau_s}f_v - f_{\tau_s}g_v}{g_\delta f_{\tau_s} - g_{\tau_s}f_\delta}, \quad \frac{d\tau_s}{dv} = \frac{f_\delta g_v - g_\delta f_v}{g_\delta f_{\tau_s} - g_{\tau_s}f_\delta}$$

Lemma 2. *In equilibrium, we have $g_\delta f_{\tau_s} - g_{\tau_s}f_\delta > 0$, and hence $\frac{d\delta_\theta}{dv} \propto g_{\tau_s}f_v - f_{\tau_s}g_v$.*

Hence, we have

$$\frac{d\delta_\theta}{d\chi} \propto g_{\tau_s}f_\chi - f_{\tau_s}g_\chi = g_{\tau_s}f_\chi = -2c\tau_s \left(-\frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} [1 - r + r\delta_\theta] \delta_\theta^2 \tau_p \right) > 0.$$

$$\frac{d\delta_\theta}{d\tau_p} \propto g_{\tau_s}f_{\tau_p} - f_{\tau_s}g_{\tau_p} = -g_{\tau_s}f_{\tau_p} < 0.$$

since $f_{\tau_p} > 0$. The results for τ_s follow by (13). The comparative statics of δ_y and δ_p are immediate from (16). Finally, we are left to prove that $\frac{\partial}{\partial \tau_p} \delta_\theta^2 \tau_p > 0$. Notice

$$\frac{d}{d\tau_p} \tau_p \delta_\theta^2 = \frac{2\delta_\theta^4}{g_\delta f_{\tau_s} - g_{\tau_s}f_\delta} \left\{ 1 - r + \frac{\sqrt{c}\delta_\theta^2 \tau_p (1 - \chi r)}{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y)} - \frac{\sqrt{c}\delta_\theta^2 \tau_p (1 - \chi)}{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y + \chi\delta_\theta^2 \tau_p)} \right\}$$

which follows from lengthy computation involving (24) and (13). Clearly, $1 - r > 0$ so it remains to show that the last two terms sum to a positive expression. This, however is immediate since $(1 - \chi r) > (1 - \chi) < \chi \iff r < 1$, which is assumed. $\square$

Proof of Proposition 2: Let us begin with δ_θ . Note that for all precision parameters, we have $g_\tau \equiv 0$, and for costs, we have $f_c \equiv 0$. For the cost parameter we have $\frac{d\delta_\theta}{dc} \propto f_{\tau_s}(\tau_s)^2 < 0$ as $f_{\tau_s} < 0$. Moreover, $\frac{d\delta_\theta}{d\tau_y} \propto g_{\tau_s}f_{\tau_y} < 0$ since $f_{\tau_y} > 0$, $\frac{d\delta_\theta}{d\tau_\theta} = \frac{\partial \delta_\theta}{\partial \tau_y} < 0$, and finally $\frac{d\delta_\theta}{dr} \propto g_{\tau_s}f_r < 0$ since $f_r \geq 0$. Using (13), we have $\frac{d\tau_s}{dv} \propto \frac{d\delta_\theta}{dv}$ for all $v \neq c$. For c , we get $\frac{d\tau_s}{dc} = \frac{1}{c} \frac{d\delta_\theta}{dc} - \tau_s c < 0$. $\square$

Let us also establish the comparative statics of δ_y and δ_p discussed in the text. The former are immediate. The latter are not globally signed. In c , we know that

$$\frac{d\delta_p}{dc} = -\frac{d\delta_\theta}{dc} - \frac{\tau_\theta + \tau_y}{1-r} \frac{1}{2\sqrt{c}},$$

so the indirect effect through δ_θ can offset the direct negative term. Concretely, we evaluate numerically and have

$$(r, \chi, \tau_p, \tau_\theta + \tau_y, c) = (0.70, 0.31, 1.63, 1.95, 0.0055) \Rightarrow \frac{d\delta_p}{dc} < 0,$$

$$(r, \chi, \tau_p, \tau_\theta + \tau_y, c) = (0.24, 0.54, 0.66, 0.36, 0.29) \Rightarrow \frac{d\delta_p}{dc} > 0.$$

For $\frac{d\delta_p}{d\tau_y}$, we show that it is negative in games of strategic complements $r \in [0, 1)$ (proof available upon request), but can be positive in games of strategic substitutes $r < 0$, as

$$(r, \chi, \tau_p, \tau_\theta, \tau_y, c) = (-2.80, 0.49, 6.72, 0.131, 0.131, 6.67) \Rightarrow \frac{d\delta_p}{d\tau_y} > 0.$$

Similarly, we can show that $\frac{d\delta_p}{dr} < 0$ for $r \in [0, 1)$ and costs close to the upper bound (15), while

$$(r, \chi, \tau_p, \tau_\theta, \tau_y, c) = (0.4, 0.68, 0.87, 0.022, 0.022, 2.79) \Rightarrow \frac{d\delta_p}{dr} > 0.$$

Theorem 3. *The loading on private information in the efficient linear action rule is the unique solution of*

$$\delta_\theta = (1 - r + r\delta_1) \tau_s^* \frac{1}{\underbrace{\left(\frac{\tau_\theta + \tau_y + \tau_p \delta_\theta^2}{\tau_\theta + \tau_y + \tau_p \delta_\theta} \right) (\tau_\theta + \tau_y + \tau_p \delta_\theta^2) + \tau_s^*}_{\text{efficiency wedge}}}. \quad (30)$$

The efficient precision of private information satisfies $\tau_s^* = \frac{\delta_\theta^*}{\sqrt{c}}$. Moreover, first-best welfare satisfies $\frac{dW^*}{d\tau_\theta} > 0$, $\frac{dW^*}{d\tau_y} > 0$, $\frac{dW^*}{d\tau_p} > 0$, and $\frac{dW^*}{dc} < 0$.

The rational equilibrium always has inefficiently low information acquisition. Sufficiently cursed agents acquire more private information than the efficient benchmark if $\tau_p > \bar{\tau}_p$, where $\bar{\tau}_p = (\tau_\theta + \tau_y) (1 - 2\delta_\theta^{\text{FC}}) (\delta_\theta^{\text{FC}})^{-3}$.

Proof of Theorem 3: Taking FOC in (21), we obtain

$$\begin{aligned} W_{\delta_\theta} &= 2(1-r) \frac{(1 - \delta_\theta - \delta_y - \delta_p)}{\tau_\theta} + 2(1-r) \frac{\delta_p^2}{\delta_\theta^3 \tau_p} - 2 \frac{\delta_\theta}{\tau_s} = 0 \\ W_{\delta_y} &= -2(1-r) \left\{ \frac{\delta_y}{\tau_y} - \frac{(1 - \delta_\theta - \delta_y - \delta_p)}{\tau_\theta} \right\} = 0 \\ W_{\delta_p} &= -2(1-r) \left\{ \frac{\delta_p}{\delta_\theta^2 \tau_p} - \frac{(1 - \delta_\theta - \delta_y - \delta_p)}{\tau_\theta} \right\} = 0 \end{aligned} \quad (31)$$

Note that the last two equations simplify to a linear system in δ_y, δ_p , which we solve to obtain

$$\delta_y = \frac{\tau_y(1 - \delta_\theta)}{\tau_\theta + \tau_y + \tau_p \delta_\theta^2}, \quad \delta_p = \frac{(1 - \delta_\theta)\tau_p \delta_\theta^2}{\tau_\theta + \tau_y + \tau_p \delta_\theta^2} \quad (32)$$

Simplifying and using the envelope condition we arrive at the defining equation for the efficient outcome

$$f^*(\delta_\theta) = (\tau_\theta + \tau_y + \delta_\theta^2 \tau_p) \left(\frac{\tau_\theta + \tau_y + \tau_p \delta_\theta^2}{\tau_\theta + \tau_y + \tau_p \delta_\theta} \right) - (1 - r) \frac{1}{\sqrt{c}} (1 - \delta_\theta) = 0$$

Lemma 3. *We have $f_{\delta_\theta}^* > 0$ for all $\delta_\theta \in (0, 1)$.*

By Lemma 3, there is a unique interior solution, as $f^*(0) = (\tau_\theta + \tau_y) - (1 - r) \frac{1}{\sqrt{c}} < 0$ by (15) and $f^*(1) = \tau_\theta + \tau_y + \tau_p > 0$ and a fortiori $f^*(\delta) > 0$ for $\delta > 1$. Reformulating $f^*(\delta_\theta) = 0$, we get the desired representation (30). Finally, plugging the efficiency conditions (32) and $\tau_s^* = \frac{\delta_\theta^*}{\sqrt{c}}$ into the welfare expression (21) we get

$$W^* := W(\delta^*, \tau_s^*) = \max_{\delta_\theta} -2\sqrt{c}\delta_\theta - \frac{(1 - r)(1 - \delta_\theta)^2}{\tau_\theta + \tau_y + \delta_\theta^2 \tau_p}. \quad (33)$$

Applying the envelope theorem, all the comparative statics follow. $\square$

Proof of Proposition 3: To establish underacquisition in the rational equilibrium, note that since $f_{\delta_\theta}^* > 0$ (Lemma 3), we know that $f^*(\delta_\theta) < 0$ implies that there is underacquisition and $f^*(\delta_\theta) > 0$ implies that there is overacquisition. Plugging the equilibrium δ_θ and using the fact that $f(\delta_\theta) = 0$

$$f^*(\delta_\theta) = (\tau_\theta + \tau_y + \delta_\theta^2 \tau_p) \left(\frac{\tau_\theta + \tau_y + \delta_\theta^2 \tau_p}{\tau_\theta + \tau_y + \delta_\theta \tau_p} - 1 \right) + \chi \frac{1}{\sqrt{c}} \frac{1}{\tau_\theta + \frac{\delta_\theta}{\sqrt{c}} + \tau_y} [1 - r + r\delta_\theta] \delta_\theta^2 \tau_p$$

Note that at $\chi = 0$, this expression is negative and hence, δ_θ is inefficiently low. As δ_θ is increasing in χ , we may exceed the efficient level for χ sufficiently large. There exists a χ with $\delta_\theta^\chi = \delta_\theta^*$ iff $f^*(\delta_\theta^{\text{FC}}) > 0$ (by $f_\delta^* > 0$). We get

$$f^*(\delta_\theta^{\text{FC}}) = \frac{\delta_\theta \tau_p}{\tau_\theta + \tau_y + \delta_\theta \tau_p} \left\{ 2(\tau_\theta + \tau_y) \delta_\theta + \delta_\theta^3 \tau_p - (\tau_\theta + \tau_y) \right\}$$

This is larger than zero iff $\tau_p \geq \frac{(\tau_\theta + \tau_y) - 2(\tau_\theta + \tau_y)\delta_\theta}{\delta_\theta^3}$. Since the cutoff is decreasing in δ_θ (and δ_θ is decreasing in χ), we obtain the sufficient bound $\bar{\tau}_p$. In particular, if $\delta_\theta^{\text{FC}} \geq \frac{1}{2}$, i.e. $\frac{\sqrt{c}(\tau_\theta + \tau_y)}{1 - r} \leq \frac{1}{2}$, the fully cursed agent always acquires more than the efficient level.

To establish the sources of inefficiency of the equilibrium, we evaluate the gradient of welfare with respect to the loadings at the equilibrium values. Plugging the equilibrium expressions (9) into W_{δ_θ} and using (13) yields

$$W_{\delta_\theta} = 2(1-r) \frac{\delta_p}{\delta_\theta^3 \tau_p} \geq 0$$

with equality for $\chi = 1$ by Corollary 2.4 and Proposition 1, establishing inefficient use unless agents are fully cursed. Plugging the equilibrium expressions for δ_y, δ_p into W_{δ_y} yields $W_{\delta_y} = 0$.

Plugging into W_{δ_p} and simplifying using $f(\delta) = 0$, we get

$$W_{\delta_p} = 2[(1-r) + r\delta_\theta] \chi \frac{\sqrt{c}}{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y)} \geq 0$$

with equality iff $\chi = 0$, establishing inefficient use if agents are fully cursed. Hence combining W_{δ_p} and W_{δ_θ} we always have inefficient use. $\square$

Proof of Proposition 4: To determine the impact of cursedness, we compute

$$\frac{dW^{EQ}}{d\chi} = \frac{\partial W^{EQ}}{\partial \delta_\theta} \frac{d\delta_\theta}{d\chi} + \frac{\partial W^{EQ}}{\partial \delta_y} \frac{d\delta_y}{d\chi} + \frac{\partial W^{EQ}}{\partial \delta_p} \frac{d\delta_p}{d\chi} = \frac{\partial W^{EQ}}{\partial \delta_\theta} \frac{d\delta_\theta}{d\chi} + \frac{\partial W^{EQ}}{\partial \delta_p} \frac{d\delta_p}{d\chi}$$

since $\frac{\partial W^{EQ}}{\partial \delta_y} = 0$. Local to $\chi = 0$, we have $\frac{\partial W^{EQ}}{\partial \delta_p} = 0$ and therefore

$$\frac{dW^{EQ}}{d\chi} \Big|_{\chi=0} = 2(1-r) \frac{\delta_p}{\delta_\theta^3 \tau_p} \cdot \frac{d\delta_\theta}{d\chi} \Big|_{\chi=0} > 0$$

Local to $\chi = 1$, we have that $\frac{\partial W^{EQ}}{\partial \delta_p} = 0$ and therefore

$$\frac{dW^{EQ}}{d\chi} \Big|_{\chi=1} = 2[(1-r) + r\delta_\theta] \frac{\sqrt{c}}{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y)} \cdot \frac{d\delta_p}{d\chi} \Big|_{\chi=1} < 0.$$

To establish quasi-concavity, note that

$$\frac{dW^{EQ}}{d\chi} = 2 \left((1-r) \frac{(1 - \delta_\theta - \sqrt{c}^{\tau_\theta + \tau_y})}{\delta_\theta^3 \tau_p} - [(1-r) + r\delta_\theta] \chi \frac{\sqrt{c}}{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y)} \right) \frac{d\delta_\theta}{d\chi}$$

defines a locus $x(\delta_\theta)$ by $\frac{dW^{EQ}}{d\chi}|_{\chi=x(\delta_\theta)} \equiv 0$ as

$$x(\delta_\theta) = \frac{(1-r) \frac{(1-\delta_\theta - \sqrt{c} \frac{\tau_\theta + \tau_y}{(1-r)})}{\delta_\theta^3 \tau_p}}{[(1-r) + r\delta_\theta] \frac{\sqrt{c}}{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y)}}.$$

An intersection between $\chi = x(\delta_\theta)$ and the equilibrium $\delta_\theta(\chi)$ read as a function of cursedness χ corresponds to an equilibrium at which equilibrium welfare takes a local maximum in χ . Such an intersection exists for any parameter combination satisfying the nondegeneracy condition (15) by the mean value theorem and the previous result about $\frac{dW^{EQ}}{d\chi}$ at the corners. If we can show that the intersection is unique, we have established quasiconcavity. Indeed, this is the case as $\frac{\partial \log x(\delta_\theta)}{\partial \delta_\theta} < 0$ in any equilibrium, as is straightforward to show.²⁰

To see the ranking in the level of welfare, note that (using the fully cursed equilibrium and (29), resp., in (21)) in the two extreme cases $\chi \in \{0, 1\}$, welfare takes the simple form

$$W^{EQ} = -\sqrt{c}(1 + \delta_\theta). \quad (34)$$

It follows from the comparative statics of δ_θ that the fully cursed equilibrium has lower welfare than the rational case. $\square$

Proof of Proposition 5: By (34), $\frac{dW}{d\tau} \propto -\frac{d\delta_\theta}{d\tau}$ for $\chi \in \{0, 1\}$ and the comparative statics w.r.t. precision parameters follow immediately from Proposition 1. For costs, note that in the rational case, direct computation yields

$$\frac{\partial W_0^{EQ}}{\partial c} = -\frac{1-r-\sqrt{c}(\tau_\theta + \tau_y) + \sqrt{c}\delta_\theta\tau_p}{\sqrt{c}(1-r) + 2c\delta_\theta\tau_p} < 0$$

²⁰At equilibrium δ_θ , we have that $x(\delta_\theta) \in (0, 1)$, and

$$\begin{aligned} \frac{\partial \log x(\delta_\theta)}{\partial \delta_\theta} &= -\frac{1}{1-\delta_\theta - \sqrt{c} \frac{\tau_\theta + \tau_y}{(1-r)}} - 3\frac{1}{\delta_\theta} - \frac{r}{1-r+r\delta_\theta} + \frac{1}{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y)} \\ &\leq -\frac{1}{\delta_p} - \frac{3}{\delta_\theta} - \frac{r}{1-r+r\delta_\theta} + \frac{1}{\delta_\theta} = -\frac{1}{\delta_p} - \frac{2}{\delta_\theta} - \frac{r}{1-r+r\delta_\theta} \end{aligned}$$

Therefore, x is decreasing: This is immediate if $r \geq 0$. If $r < 0$, we can write $r = -|r|$ and note that $-\frac{r}{1-r+r\delta_\theta} = \frac{|r|}{1+|r|(1-\delta_\theta)} \leq \frac{1}{1-\delta_\theta}$ and hence

$$-\frac{1}{\delta_p} - \frac{2}{\delta_\theta} - \frac{r}{1-r+r\delta_\theta} \leq \frac{1}{1-\delta_\theta} - \frac{1}{1-\delta_\theta - \sqrt{c} \frac{\tau_\theta + \tau_y}{(1-r)}} - \frac{2}{\delta_\theta} \leq -\frac{2}{\delta_\theta} < 0.$$

which is negative by the parameter condition (15). In the fully cursed case, note that

$$\frac{\partial W_1^{\text{EQ}}}{\partial c} = -\frac{1}{\sqrt{c}} \left(\frac{(1-r) - \sqrt{c}(\tau_\theta + \tau_y)}{1-r} \right) = -\frac{\delta_\theta}{\sqrt{c}} < 0$$

For $\tau_p = 0$, the rational and (partially) cursed equilibria coincide, hence the above comparative statics prevail for all χ , and by continuity, this extends to small but interior τ_p . The paradoxical welfare results emerge instead for large τ_p . To make them apparent we consider the limit case $\tau_p \rightarrow \infty$. By continuity, the following comparative statics hold for sufficiently large τ_p .

Lemma 4. *The limit welfare (57) is*

- *Decreasing in τ_θ, τ_y if and only if $r > 0$ and $\chi \leq \frac{2r-1}{r}$.*
- *Decreasing in costs, unless $r < 0$, when higher costs increase welfare for intermediate χ .*

It remains to be shown that welfare is always increasing in τ_p . Differentiating and using the linear dependence $\frac{\partial \delta_p}{\partial \delta_\theta} = -1$, we get

$$\frac{dW}{d\tau_p} = (1-r) \frac{\delta_p^2}{\delta_\theta^2 \tau_p^2} + \frac{(1-r)}{\tau_p} \left(\frac{2\delta_p}{\delta_\theta^2} + \frac{2\delta_p^2}{\delta_\theta^3} \right) \frac{d\delta_\theta}{d\tau_p} - 2\sqrt{c} \frac{d\delta_\theta}{d\tau_p}. \quad (35)$$

Rewriting (29) as

$$0 = \tilde{f}(\delta_\theta) = \underbrace{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y) - (1-r) - r\delta_\theta + \tau_p \tilde{f}_{\tau_p}}_{=-(1-r)\delta_p}$$

implies that $\tilde{f}_{\tau_p} = \frac{(1-r)\delta_p}{\tau_p}$; also recall that by implicit differentiation $\frac{d\delta_\theta}{d\tau_p} = -\frac{\tilde{f}_{\tau_p}}{\tilde{f}_{\delta_\theta}}$. Substituting these expressions into (35) and noticing that $\frac{(1-r)}{\tau_p} \left(\frac{2\delta_p}{\delta_\theta^2} + \frac{2\delta_p^2}{\delta_\theta^3} \right) = \frac{2\tilde{f}_{\tau_p}}{\delta_\theta^2} \left(1 + \frac{\delta_p}{\delta_\theta} \right)$ we obtain

$$\frac{dW}{d\tau_p} = \frac{\tilde{f}_{\tau_p}^2}{(1-r)\delta_\theta^2} - \left[\frac{2\tilde{f}_{\tau_p}}{\delta_\theta^2} \left(1 + \frac{\delta_p}{\delta_\theta} \right) - 2\sqrt{c} \right] \frac{\tilde{f}_{\tau_p}}{\tilde{f}_{\delta_\theta}}. \quad (36)$$

Denote $A := 1 - \chi \frac{(1-r)+r\delta_\theta}{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y)}$ and notice that, from direct differentiation, $\tilde{f}_{\tau_p} = \sqrt{c}\delta_\theta^2 A$. Since $\tilde{f}_{\tau_p} > 0$ then also $A > 0$; moreover $A < 1$ by Corollary 1. Substituting this expression yields

$$\frac{dW}{d\tau_p} = \frac{cA^2\delta_\theta^2}{(1-r)} - \left[2\sqrt{c}A \left(1 + \frac{\delta_p}{\delta_\theta} \right) - 2\sqrt{c} \right] \frac{\sqrt{c}\delta_\theta^2 A}{\tilde{f}_{\delta_\theta}} \quad (37)$$

Moreover, differentiating $\tilde{f}(\delta_\theta) = -(1-r)\delta_p + \tilde{f}_{\tau_p}\tau_p$ with respect to δ_θ we obtain that

$$\tilde{f}_{\delta_\theta} = -(1-r)\frac{\partial}{\partial\delta_\theta}\delta_p + \frac{\partial}{\partial\delta_\theta}\tilde{f}_{\tau_p}\tau_p = (1-r) + \tau_p\sqrt{c}\delta_\theta(2A + \delta_\theta A_{\delta_\theta})$$

Using that $\sqrt{c}\delta_\theta^2 A = \tilde{f}_{\tau_p} = \frac{(1-r)\delta_p}{\tau_p}$, we obtain $\frac{1}{1-r} = \frac{1+2\frac{\delta_p}{\delta_\theta}+\delta_p\frac{A_{\delta_\theta}}{A}}{\tilde{f}_{\delta_\theta}}$ which we substitute to get

$$\frac{dW}{d\tau_p} = \frac{cA^2\delta_\theta^2}{(1-r)} - \left[2A\left(1 + \frac{\delta_p}{\delta_\theta}\right) - 2 \right] \frac{c\delta_\theta^2 A}{\tilde{f}_{\delta_\theta}} = \frac{cA\delta_\theta^2}{\tilde{f}_{\delta_\theta}} (2 - A + \delta_p A_{\delta_\theta}) \quad (38)$$

Since $\tilde{f}_{\delta_\theta} > 0$ and $A > 0$, the sign is determined by $2 - A + \delta_p A_{\delta_\theta}$. Notice that

$$A_{\delta_\theta} = -\chi \frac{\partial}{\partial\delta_\theta} \left(\frac{1-r+r\delta_\theta}{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y)} \right) = \chi \frac{(1-r) - r\sqrt{c}(\tau_\theta + \tau_y)}{(\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y))^2} > 0.$$

The last inequality follows from $(1-r) - r\sqrt{c}(\tau_\theta + \tau_y)$; if $r < 0$ it is obvious, while for $r \in (0, 1)$ it is implied by the non-triviality condition $\sqrt{c} < \frac{1-r}{\tau_\theta + \tau_y}$. Since $\delta_p > 0$, we have $\delta_p A_{\delta_\theta} \geq 0$, and since $0 < A < 1$ we have $2 - A > 1$. Therefore $2 - A + \delta_p A_{\delta_\theta} > 0$, establishing that $\frac{dW}{d\tau_p} > 0$. $\square$

B Shrewd Agent: Behaviour and Policy

Although their welfare increases with transparency, cursed agents fail to reap its full benefits. How does an agent who is able to extract all the information from the environment – such as proverbial smart money in financial markets – interact with a cursed crowd? Could it be beneficial to act in an environment with less rational agents? We address these questions by studying the behaviour and welfare of a *shrewd agent*: a fully rational, atomistic agent in the model who understands its structure and is aware that all other agents (the *cursed crowd*) are χ -cursed.

B.1 Best Response and Information Acquisition

We continue to denote the precision of information acquired by the cursed crowd as τ_s and denote the precision acquired by the shrewd agent as τ_s^R . The shrewd agent takes the equilibrium loadings (and information acquisition) of the cursed crowd as given, and chooses both how much private information to acquire as well as the coefficients in his linear action rule $a_i^R = \alpha_s^R s_i + \alpha_y^R y + \alpha_p^R \hat{p}$. Formally, he solves

$$\max_{\alpha^R, \tau_s^R} W(\alpha^R, \delta, \tau_s^R) \quad (39)$$

Best responding to the equilibrium in the cursed crowd, the individual loadings of the shrewd agent α^R will differ from δ whenever $\chi > 0$.

Proposition 6. *The action rule of the shrewd agent is*

$$\alpha_s^R = \frac{(1 - r + r\delta_\theta) \tau_s^R}{\tau_\theta + \tau_y + \tau_s^R + \delta_\theta^2 \tau_p} \quad (40)$$

$$\alpha_y^R = \frac{(1 - r + r\delta_\theta) \tau_y}{\tau_\theta + \tau_y + \tau_s^R + \delta_\theta^2 \tau_p} + r\sqrt{c} \frac{\tau_y}{1 - r} \quad (41)$$

$$\alpha_p^R = \frac{(1 - r + r\delta_\theta) \delta_\theta^2 \tau_p}{\tau_\theta + \tau_y + \tau_s^R + \delta_\theta^2 \tau_p} + r \left(1 - \delta_\theta - \sqrt{c} \frac{\tau_\theta + \tau_y}{1 - r} \right) \quad (42)$$

where τ_s^R solves

$$\tau_s^R = \frac{\alpha_s^R}{\sqrt{c}} \quad (43)$$

with complementary slackness ensuring $\tau_s^R \geq 0$ when required.

There is again a tight connection between the use of private information, α_s^R , and its acquisition through the envelope condition (43). Denote the total precision available to the cursed agents by $\tau_\Sigma := \tau_\theta + \tau_y + \tau_s + \delta_\theta^2 \tau_p$ and that available to the rational agent by $\tau_\Sigma^R := \tau_\theta + \tau_y + \tau_s^R + \delta_\theta^2 \tau_p$. We obtain an equation linking the information acquired by the shrewd agent and the cursed crowd

$$\frac{\tau_\Sigma}{\tau_\Sigma^R} = 1 + \frac{\chi \delta_\theta^2 \tau_p}{\tau_\theta + \tau_y + \tau_s}. \quad (44)$$

The shrewd agent acquires less information. Compared to the cursed crowd, he can substitute it with a better comprehension of aggregative information. If the crowd is fully cursed, then (44) simplifies to

$$\tau_s^R = \tau_s - \delta_\theta^2 \tau_p \quad (45)$$

that is, the shrewd agent exactly offsets the information he can glean from the aggregative signal and his precision is equal to the precision *perceived* by the crowd.

Clearly, equation (45) holds only if it delivers a positive τ_s^R . Otherwise, the shrewd agent will choose $\tau_s^R = 0$ as he is already satiated with the information he can infer from the aggregative signal. With a fully cursed crowd, this always happens with sufficiently large transparency since both τ_s and δ_θ are unresponsive to τ_p . In that case, transparency only serves as a cost-saving device for the shrewd agent. The shrewd agent continues to free-ride on the crowd's use of private information even at interior levels of cursedness.

Proposition 7. *The information acquired by the shrewd agent, τ_s^R , satisfies:*

1. *It is bounded by the precision acquired by the cursed crowd $\tau_s^R \leq \tau_s$. It grows without bounds as costs vanish but can be zero even when $\tau_s > 0$.*
2. *If τ_p is sufficiently large, then it is*
 - (i) *nonmonotonic in prior and public precision, τ_θ and τ_y , possibly with an interior activity region; and*
 - (ii) *nonmonotonic in costs c , possibly with an interior inactivity region.*
3. *It has ambiguous comparative statics w.r.t. χ ,*

$$\frac{d\tau_s^R}{d\chi} \propto r - 2\sqrt{c}\delta_\theta\tau_p.$$

The shrewd agent acquires a strictly positive amount of information if and only if $\tau_\theta + \tau_y \in \left(\frac{1-r}{\sqrt{c}}\left(1 - \frac{1}{\tau_p\sqrt{c}}\right), \frac{1-r}{\sqrt{c}}\right)$. Therefore, there is an inactivity region whenever $\tau_p > \frac{1}{\sqrt{c}}$. In that case, the shrewd agent acquires private information only if public information is sufficiently precise. The rationale is as follows: If public fundamental information is noisy, the cursed crowd will acquire and use a lot of private information; since he can be parasitic on this information, the rational agent has no incentive to acquire information himself. As public information becomes more abundant, however, there is less information acquisition and use by the crowd. The aggregative source dries up and the shrewd agent needs to supplement it with private information acquisition. Finally, the upper bound on $\tau_\theta + \tau_y$ for the existence of a nontrivial equilibrium is the same for both classes of agents. In the trivial equilibrium $\delta_\theta = \tau_s = 0$, the shrewd agent cannot utilize his advantage in understanding the aggregative source since it is uninformative: he behaves identically to the crowd.

An immediate consequence of this inactivity region is that τ_s^R is nonmonotonic in τ_y . This contrasts with the unambiguously signed comparative statics for τ_s (Proposition 1). Similarly, the effect of information acquisition costs on τ_s^R is nonmonotonic and we can have an inactivity region (see Figure 7). Again, a change in parameters affects both the availability of aggregative information provided by the cursed crowd and the shrewd agent's demand for information overall.

The impact of cursedness on the precision of information acquired by the shrewd agent depends on the nature of strategic interactions (see Figure 7). Take as a benchmark the case of $r = 0$, i.e., all agents simply try to guess the true state, and all strategic interaction comes from the precision of aggregative information. In this case, τ_s^R is decreasing in χ as more cursed agents acquire and disseminate more information. Strategic substitutes increase this effect: As the crowd becomes more cursed, δ_θ

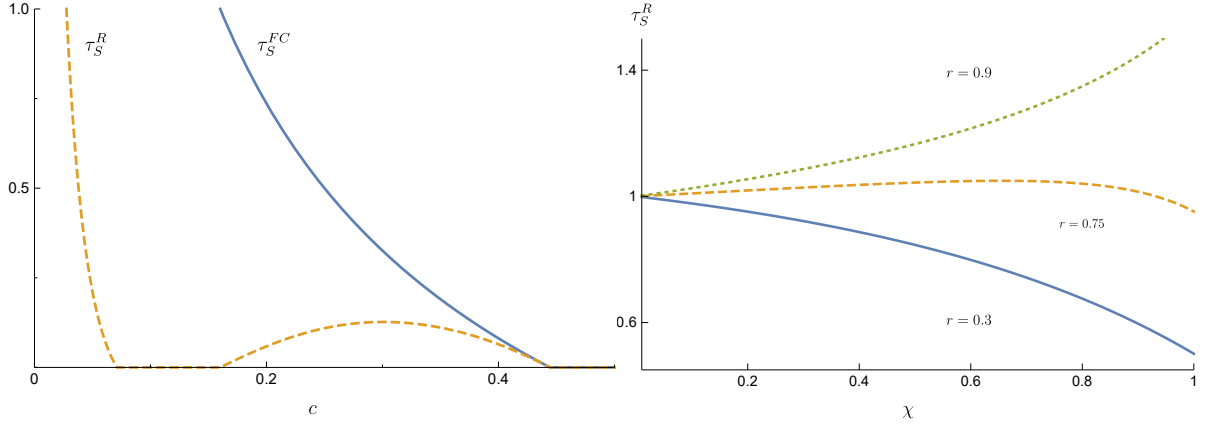


Figure 7: τ_s and τ_s^R as a function of c (left). τ_s^R (normalized to 1 at $\chi = 0$) as a function of χ for $r \in \{0.3, 0.75, 0.9\}$ (right).

increases, which reduces the desire to match the state and therefore the value of private information. With complements, the opposite is the case: A higher δ_θ increases the desire to match θ and therefore – if this motive is sufficiently strong – information acquisition.

B.2 Welfare

Let W_χ^R denote the welfare of the shrewd agent facing an equilibrium δ^χ . Then,

$$W_\chi^{\text{EQ}} = W(\delta^\chi, \delta^\chi, \tau_s^\chi) \leq \max_{\alpha, \tau_s^R} W(\alpha, \delta^\chi, \tau_s^R) = W_\chi^R \quad (46)$$

As he comprehends his information environment, he always obtains a higher welfare than the cursed crowd. The inequality is strict if $\chi > 0$.

We now ask whether the shrewd agent benefits from an increase in the cursedness of the crowd. This is the case for the first modicum of cursedness since for small positive ϵ

$$W_\epsilon^R > W_\epsilon^{\text{EQ}} > W_0^{\text{EQ}} = W_0^R. \quad (47)$$

The central inequality follows since in this region “cursedness is bliss” (Proposition 4).²¹ In a highly cursed environment, however, the impact of cursedness depends on nature of the strategic interaction.

Proposition 8. *Suppose parameters are such that $\tau_s^R > 0$. If $r \leq 0$, then $\left. \frac{dW_\chi^R}{d\chi} \right|_{\chi=1} \geq 0$.*

However, for r sufficiently large, $\left. \frac{dW_\chi^R}{d\chi} \right|_{\chi=1} \leq 0$.

²¹Indeed, the information spillover can be strong enough to make the shrewd agent in the cursed world better off than first-best welfare (as can be checked for $r = 0$, $\tau_\theta = \tau_y = 0.1$, $\tau_p = 0.19$, $c = 0.03$, where we have $W_1^R > W^*$). By continuity, this holds for an open set of parameters.

If there are strategic substitutes, the shrewd agent always benefits from increasing the cursedness of the crowd: not only does he free-ride on aggregative information, but the crowd's over-reliance on the private signal helps him anti-coordinate. In the presence of complementarities, however, informational free-riding and the lack of coordination implied by cursed information misuse have opposing effects. While the shrewd agent can learn the state more precisely, his action has to follow the behaviour of the less informed crowd. The latter effect can be overwhelming close to $\chi = 1$, so he would prefer an interior level of cursedness.

We conclude this section by studying the impact of precision and cost parameters on W_χ^R . If the crowd is close to rational, then policies have an impact similar to that in the rational equilibrium. We therefore focus our analysis on the other extreme case and evaluate the welfare of the shrewd agent facing a fully cursed crowd. By continuity, the results extend to a sufficiently cursed environment.

We now discuss the effect of information policies on the welfare of a shrewd agent that plays against a fully cursed crowd. First, notice that W_1^{EQ} is independent of τ_p as fully cursed agents do not respond to transparency. Therefore, higher transparency only affects the shrewd agent by providing a more precise aggregative signal, which is clearly beneficial. Recall that more public information τ_y and lower cost c are always beneficial for the cursed crowd (Proposition 5). The shrewd agent, however, suffers from the crowding out effect, as higher τ_y decreases information acquisition and dissemination by the cursed crowd. This is especially harmful if he is largely relying on this source of information, leading to the negative welfare impact when τ_s^R is small.

Higher c can be beneficial for the shrewd agent if τ_s^R is zero, i.e. if there is no direct effect of higher costs. Suppose that this is the case, that aggregate information is relatively abundant, and that $r < 0$ (strategic substitutes). As c increases, cursed agents rely more on y which makes it easier for the shrewd agent to anti-coordinate. This action externality can dominate the harm from reduced information dissemination. When τ_s^R is positive, by contrast, the direct effect always dominates, yielding the natural comparative static. Summarizing the discussion,

Proposition 9. *The welfare of a shrewd agent in a fully cursed population, W_1^R , has the following properties.*

1. *It is strictly increasing in τ_p .*
2. *It has ambiguous comparative statics with respect to τ_y . In particular,*
 - *At the boundary of the activity region, (i.e. $\tau_y = \frac{(1-r)\left(1 - \frac{1}{\tau_p \sqrt{c}}\right)}{\sqrt{c}} - \tau_\theta$), we have $\frac{dW_1^R}{d\tau_y} < 0$,*
 - *for τ_y large (i.e. local to the nontriviality limit $\tau_y = \frac{(1-r)}{\sqrt{c}} - \tau_\theta$), we have $\frac{dW_1^R}{d\tau_y} > 0$.*

3. It is decreasing in c whenever $\tau_s^R > 0$, but it has ambiguous comparative statics with respect to c if $\tau_s^R = 0$. In particular,

- If $\tau_p \geq \frac{1}{\sqrt{c}}$ and r sufficiently negative, then $\frac{dW_I^R}{dc} > 0$.

The comparison between Proposition 5 and 9 highlights qualitative differences in the impact of policy on the welfare of cursed and shrewd agents. Transparency leaves the welfare of the cursed crowd unaffected but has a strictly positive (and large) impact for the shrewd. If both types affected the aggregate outcome, this could easily turn into a redistribution result, suggesting that transparency can function as an elitist policy that gives an advantage to sophisticated agents who are able to understand and utilize aggregative information. For public information and lower cost this trade-off is already apparent in the present results.²² A natural avenue for studying these questions further would be to extend our model to include true cognitive heterogeneity, featuring several non-atomistic groups with different levels of cursedness, all affecting the aggregate outcomes. Although the linear structure of the model makes action aggregation straightforward, the correlation between information use and acquisition affects the aggregate outcome and introduces nonlinearity. The analysis of such cognitive heterogeneity is therefore beyond the scope of this paper.

References

- Amador, Manuel and Pierre-Olivier Weill**, “Learning from Prices: Public Communication and Welfare,” *Journal of Political Economy*, 2010, 118 (5), 866–907.
- **and —**, “Learning from Private and Public Observations of Others’ Actions,” *Journal of Economic Theory*, May 2012, 147 (3), 910–940.
- Angeletos, George-Marios and Alessandro Pavan**, “Efficient Use of Information and Social Value of Information,” *Econometrica*, 2007, 75 (4), 1103–1142.
- **and Chen Lian**, “Forward Guidance without Common Knowledge,” *American Economic Review*, September 2018, 108 (9), 2477–2512.
- **and Iván Werning**, “Crises and Prices: Information Aggregation, Multiplicity, and Volatility,” *American Economic Review*, December 2006, 96 (5), 1720–1736.
- Angeletos, Marios and Jennifer La’O**, “Noisy Business Cycles,” *NBER Macroeconomics Annual* 2009, April 2010, 24, 319–378.
- Angrisani, Marco, Antonio Guarino, Philippe Jehiel, and Toru Kitagawa**, “Information Redundancy Neglect versus Overconfidence: A Social Learning Experiment,” *American Economic Journal: Microeconomics*, August 2021, 13 (3), 163–197.

²²This conflict of interest between experts and unsophisticated actors casts doubt on the role of expert lobbying as a source of information on the impact of such policies.

- Baeriswyl, Romain, Camille Cornand, and Bruno Ziliotto**, “Observing and Shaping the Market: The Dilemma of Central Banks,” *Journal of Money, Credit and Banking*, December 2020, 52 (8), 1973–2005.
- Barberis, Nicholas**, “Psychology-Based Models of Asset Prices and Trading Volume,” in “Handbook of Behavioral Economics: Applications and Foundations 1,” Vol. 1, Elsevier, 2018, pp. 79–175.
- Bayona, Anna**, “The Social Value of Information with an Endogenous Public Signal,” *Economic Theory*, December 2018, 66 (4), 1059–1087.
- **and Carolina Manzano**, “Competition in Schedules with Cursed Traders,” *Journal of Economic Theory*, December 2024, 222, 105935.
- Bohren, Aislinn and Daniel Hauser**, “The Behavioral Foundations of Model Misspecification,” 2023.
- Cavallo, Alberto, Guillermo Cruces, and Ricardo Perez-Truglia**, “Inflation Expectations, Learning, and Supermarket Prices: Evidence from Survey Experiments,” *American Economic Journal: Macroeconomics*, July 2017, 9 (3), 1–35.
- Charness, Gary and Dan Levin**, “The Origin of the Winner’s Curse: A Laboratory Study,” *American Economic Journal: Microeconomics*, February 2009, 1 (1), 207–236.
- Cipriani, Marco and Antonio Guarino**, “Herd Behavior in a Laboratory Financial Market,” *American Economic Review*, November 2005, 95 (5), 1427–1443.
- Cohen, Shani**, “What You Can’t See: Contingent Thinking Failures in Dynamic Markets,” 2026.
- **and Shengwu Li**, “Sequential Cursed Equilibrium,” February 2023.
- Colombo, L., G. Femminis, and A. Pavan**, “Information Acquisition and Welfare,” *The Review of Economic Studies*, October 2014, 81 (4), 1438–1483.
- Conlon, John J, Malavika Mani, Gautam Rao, Matthew Ridley, and Frank Schilbach**, “Not Learning from Others,” March 2026.
- Diamond, Douglas W. and Robert E. Verrecchia**, “Information Aggregation in a Noisy Rational Expectations Economy,” *Journal of Financial Economics*, September 1981, 9 (3), 221–235.
- Drehmann, Mathias, Jörg Oechssler, and Andreas Roider**, “Herding and Contrarian Behavior in Financial Markets: An Internet Experiment,” *American Economic Review*, December 2005, 95 (5), 1403–1426.
- Esponda, Ignacio and Emanuel Vespa**, “Hypothetical Thinking and Information Extraction in the Laboratory,” *American Economic Journal: Microeconomics*, November 2014, 6 (4), 180–202.
- Eyster, Erik**, “Errors in Strategic Reasoning,” in “Handbook of Behavioral Economics: Applications and Foundations 1,” Vol. 2, Elsevier, 2019, pp. 187–259.
- **and Matthew Rabin**, “Cursed Equilibrium,” *Econometrica*, September 2005, 73 (5), 1623–1672.

- , —, and **Dimitri Vayanos**, “Financial Markets Where Traders Neglect the Informational Content of Prices,” *The Journal of Finance*, 2019, 74 (1), 371–399.
- Fong, Meng-Jhang, Po-Hsuan Lin, and Thomas R. Palfrey**, “Cursed Sequential Equilibrium,” April 2023.
- Goeree, Jacob K. and Leeat Yariv**, “Conformity in the Lab,” *Journal of the Economic Science Association*, July 2015, 1 (1), 15–28.
- Grossman, Sanford J. and Joseph E. Stiglitz**, “On the Impossibility of Informationally Efficient Markets,” *The American Economic Review*, 1980, 70 (3), 393–408.
- Hebert, Benjamin and Jennifer La’O**, “Information Acquisition, Efficiency, and Non-Fundamental Volatility,” *Journal of Political Economy*, October 2023, 131 (10), 2666–2723.
- Hellwig, Christian and Laura Veldkamp**, “Knowing What Others Know: Coordination Motives in Information Acquisition,” *The Review of Economic Studies*, January 2009, 76 (1), 223–251.
- Ivanov, Asen, Dan Levin, and Muriel Niederle**, “Can Relaxation of Beliefs Rationalize the Winner’s Curse?: An Experimental Study,” *Econometrica*, 2010, 78 (4), 1435–1452.
- Kagel, John H. and Dan Levin**, *Common Value Auctions and the Winner’s Curse*, Princeton, N.J: Princeton University Press, 2002.
- Kool, Clemens, Menno Middeldorp, and Stephanie Rosenkranz**, “Central Bank Transparency and the Crowding Out of Private Information in Financial Markets,” *Journal of Money, Credit and Banking*, June 2011, 43 (4), 765–774.
- Kyle, Albert S.**, “Continuous Auctions and Insider Trading,” *Econometrica*, November 1985, 53 (6), 1315–1335.
- Lucas, Robert E.**, “Expectations and the Neutrality of Money,” *Journal of Economic Theory*, April 1972, 4 (2), 103–124.
- Minelli, E. and H. Polemarchakis**, “Information at Equilibrium,” *Economic Theory*, 2003, 21 (2/3), 573–584.
- Morris, Stephen and Hyun Song Shin**, “Social Value of Public Information,” *American Economic Review*, November 2002, 92 (5), 1521–1534.
- and —, “Central Bank Transparency and the Signal Value of Prices,” *Brookings Papers on Economic Activity*, 2005, 36 (2), 1–66.
- Ngangoué, M. Kathleen and Georg Weizsäcker**, “Learning from Unrealized versus Realized Prices,” *AEJ: Microeconomics*, May 2021, 13 (2), 174–201.
- Rostek, Marzena and Marek Weretka**, “Price Inference in Small Markets,” *Econometrica*, 2012, 80 (2), 687–711.
- Shadmehr, Mehdi, Sepehr Shahshahani, and Charles Cameron**, “Coordination and Innovation in Judiciaries: Correct Law versus Consistent Law,” *Quarterly Journal of Political Science*, January 2022, 17 (1), 61–89.
- Svensson, Lars E. O.**, “Social Value of Public Information: Comment: Morris and Shin

- (2002) Is Actually Pro-Transparency, Not Con,” *American Economic Review*, March 2006, 96 (1), 448–452.
- Trevino, Isabel**, “Informational Channels of Financial Contagion,” *Econometrica*, 2020, 88 (1), 297–335.
- Uhlig, Harald**, “A Law of Large Numbers for Large Economies,” *Economic Theory*, February 1996, 8 (1), 41–50.
- Ui, Takashi and Yasunori Yoshizawa**, “Characterizing Social Value of Information,” *Journal of Economic Theory*, July 2015, 158, 507–535.
- Vives, Xavier**, “Aggregation of Information in Large Cournot Markets,” *Econometrica*, 1988, 56 (4), 851–876.
- , *Information and Learning in Markets: The Impact of Market Microstructure*, Princeton: Princeton University Press, 2008.
- , “Strategic Supply Function Competition With Private Information,” *Econometrica*, 2011, 79 (6), 1919–1966.
- , “Endogenous Public Information and Welfare in Market Games,” *The Review of Economic Studies*, April 2017, 84 (2), 935–963.
- Weizsäcker, Georg**, “Do We Follow Others When We Should? A Simple Test of Rational Expectations,” *American Economic Review*, December 2010, 100 (5), 2340–2360.
- Wong, Jacob**, “Information Acquisition, Dissemination, and Transparency of Monetary Policy,” *The Canadian Journal of Economics*, 2008, 41 (1), 46–79.
- Woodford, Michael**, “Central Bank Communication and Policy Effectiveness,” *Proceedings - Economic Policy Symposium - Jackson Hole*, 2005, (Aug), 399–474.

C Online Appendix

Acquisition, (Mis)use and Dissemination of Information: The Blessing of Cursedness and Transparency

The online appendix is structured as follows:

Appendix C.1 formally develops the mixture distribution replicating cursed interim beliefs discussed in Section 6.2.

Appendix C.2 states and proves results about the economy with exogenous τ_s .

Appendix C.3 proves the results of Appendix B (shrewd agent).

Appendix C.4 proves technical lemmas stated and used in Appendix A (proofs).

C.1 Derivation of the quasi-Bayesian Mixture Distribution

Formally, the interim χ -cursed posterior (Eyster and Rabin, 2005) is given by the mixture distribution

$$\theta|s_i, y, \hat{p} \sim \mu_\chi := \chi \mathcal{N}\left(\frac{\tau_y y + \tau_s s_i}{\tau_\theta + \tau_y + \tau_s}, \frac{1}{\tau_\theta + \tau_y + \tau_s}\right) + (1-\chi) \mathcal{N}\left(\frac{\tau_y y + \tau_s s_i + \delta_\theta^2 \tau_p \hat{p}}{\tau_\theta + \tau_y + \tau_s + \delta_\theta^2 \tau_p}, \frac{1}{\tau_\theta + \tau_y + \tau_s + \delta_\theta^2 \tau_p}\right).$$

Interim welfare is then given by

$$w(s_i, y, \hat{p}) = \mathbb{E}_{\mu_\chi} \left[-(1-r)(a_i - \theta)^2 - r(a_i - \bar{a})^2 \right] = a_i^2 - (1-r) \left(\mathbb{V}_{\mu_\chi}[\theta] + \mathbb{E}_{\mu_\chi}^2[\theta] \right) - r \left(\mathbb{V}_{\mu_\chi}[\bar{a}] + \mathbb{E}_{\mu_\chi}^2[\bar{a}] \right)$$

where a_i denotes the χ -cursed optimal action (4) which is measurable w.r.t. $(s_i, y, \hat{p})$. To estimate the value of private information ex-ante, the agent requires a joint prior over the signals (that also takes into account their correlation through θ). Note that these beliefs over signal realizations do not depend on cursedness. Cursedness only affects the inference from the aggregative signal to the state, not the perceived distribution over signals having integrated out the state.²³ The prior distribution is therefore given by

$$\begin{pmatrix} s_i \\ y \\ \hat{p} \end{pmatrix} \sim \mu(\tau_s) := \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{1}{\tau_\theta} + \frac{1}{\tau_s} & \frac{1}{\tau_\theta} & \frac{1}{\tau_\theta} \\ \frac{1}{\tau_\theta} & \frac{1}{\tau_\theta} + \frac{1}{\tau_y} & \frac{1}{\tau_\theta} \\ \frac{1}{\tau_\theta} & \frac{1}{\tau_\theta} & \frac{1}{\tau_\theta} + \frac{1}{\delta_\theta^2 \tau_p} \end{pmatrix} \right) \quad (48)$$

²³To see this, note that a fully cursed agent perceives the rescaled aggregative signal to be $\hat{p} = \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} s_i + \frac{\tau_y}{\tau_\theta + \tau_y + \tau_s} y + \text{noise} + \frac{1}{\delta_\theta} z_p$, where the noise is uncorrelated with the other signals and the state. This is because information about θ affects the agents beliefs about p even though he doesn't update vice versa. It is easy to check from this representation that the covariance of p with the other signals is unaffected by cursedness.

The **quasi-Bayesian** level of information acquisition solves

$$\max_{\tau_s} \mathbb{E}_{\mu(\tau_s)} [w(s_i, y, p)] - c\tau_s,$$

where the expectation is taken under the measure (48).

From an ex-ante perspective, the fully cursed prior is

$$\begin{pmatrix} \theta \\ s_i \\ y \\ \widehat{p} \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{1}{\tau_\theta} & \frac{1}{\tau_\theta} & \frac{1}{\tau_\theta} & \frac{\tau_y + \tau_s}{\tau_\theta + \tau_y + \tau_s} \frac{1}{\tau_\theta} \\ \frac{1}{\tau_\theta} & \frac{1}{\tau_\theta} + \frac{1}{\tau_s} & \frac{1}{\tau_\theta} & \frac{1}{\tau_\theta} \\ \frac{1}{\tau_\theta} & \frac{1}{\tau_\theta} & \frac{1}{\tau_\theta} + \frac{1}{\tau_y} & \frac{1}{\tau_\theta} \\ \frac{\tau_y + \tau_s}{\tau_\theta + \tau_y + \tau_s} \frac{1}{\tau_\theta} & \frac{1}{\tau_\theta} & \frac{1}{\tau_\theta} & \frac{1}{\tau_\theta} + \frac{1}{\delta_\theta^2 \tau_p} \end{pmatrix} \right)$$

where we see how the correlation between $\hat{p}$ and the state depends on the *individually chosen* precision of the private signal. This is required to ensure that θ and $\hat{p}$ are independent conditional on s_i and y , while the marginal distribution of $\hat{p}$ conditional on s_i, y is correctly specified.

C.2 Model With Exogenous Private Information τ_s

We start by analysing the model with exogenous private information, some results from which are cited in Claim 1 and the surrounding discussion to clarify the impact of endogenous information acquisition. We first establish a helpful lemma.

Lemma 5. *In equilibrium, we have $f_\delta > 0$.*

Proof. Compute

$$\begin{aligned} f_\delta &= \tau_\theta + \tau_y + 3\delta_\theta^2 \tau_p - \chi \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} [2 + r(3\delta_\theta - 2)] \delta_\theta \tau_p + (1 - r) \tau_s \\ &= \left(\tau_\theta + \tau_y + 3\delta_\theta^2 \tau_p \left[1 - \chi \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} \left[\frac{1 + r(\delta_\theta - 1)}{\delta_\theta} \right] \right] \right) + \chi \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} [1 - r] \delta_\theta \tau_p + (1 - r) \tau_s \\ &= \left(\tau_\theta + \tau_y + 3\delta_\theta^2 \tau_p \left[1 - \frac{\chi \tau_\theta + \chi \tau_y + \chi \tau_s + \chi \delta_\theta^2 \tau_p}{\tau_\theta + \tau_y + \tau_s + \chi \delta_\theta^2 \tau_p} \right] \right) + \chi \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} [1 - r] \delta_\theta \tau_p + (1 - r) \tau_s > 0 \end{aligned}$$

where in the final step we used (24) in the transformation:

$$1 - \chi \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} \left[\frac{1 - r + r\delta_\theta}{\delta_\theta} \right] = 1 - \frac{\chi \tau_\theta + \chi \tau_y + \chi \tau_s + \chi \delta_\theta^2 \tau_p}{\tau_\theta + \tau_y + \tau_s + \chi \delta_\theta^2 \tau_p} \geq 0. \quad \square$$

Proposition 10 (Proposition 1 w/o information acquisition). *The comparative statics of*

the model with exogenous τ_s are given by

$$\frac{\partial \delta_\theta}{\partial \chi} \geq 0, \quad \frac{\partial \delta_y}{\partial \chi} \geq 0, \quad \frac{\partial \delta_p}{\partial \chi} \leq 0, \quad (49)$$

$$\frac{\partial \delta_\theta}{\partial \tau_p} \leq 0, \quad \frac{\partial \delta_y}{\partial \tau_p} \leq 0, \quad \frac{\partial \delta_p}{\partial \tau_p} \geq 0, \quad (50)$$

all inequalities being strict if $\tau_p \neq 0$ and $\chi \neq 1$. Furthermore,

$$\frac{\partial}{\partial \tau_p} \delta_\theta^2 \tau_p > 0. \quad (51)$$

Proof. χ comparative statics: By implicit differentiation

$$\frac{d\delta_\theta}{d\chi} = -\frac{f_\chi}{f_\delta} \propto \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} [(1-r) + r\delta_\theta] \delta_\theta^2 \tau_p \geq 0$$

and from (9), it is immediate that $\frac{d\delta_y}{d\chi} \propto \frac{d\delta_\theta}{d\chi} \geq 0$ and $\frac{d\delta_p}{d\chi} \propto -\frac{d\delta_\theta}{d\chi} \leq 0$.

τ_p comparative statics: Again, $\frac{d\delta_\theta}{d\tau_p} \propto -f_{\tau_p}$ and using (24), we have

$$f_{\tau_p} = \delta_\theta^3 - \chi \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} [(1-r) + r\delta_\theta] \delta_\theta^2 = \delta_\theta^3 \left(1 - \chi \frac{\tau_\theta + \tau_y + \tau_s + \delta_\theta^2 \tau_p}{\tau_\theta + \tau_y + \tau_s + \chi \delta_\theta^2 \tau_p} \right) > 0$$

From (9), the comparative statics are immediate. Finally, we have

$$\begin{aligned} \frac{d\delta_\theta^2 \tau_p}{d\tau_p} &= 2\delta_\theta \tau_p \frac{d\delta_\theta}{d\tau_p} + \delta_\theta^2 \\ &= -2\delta_\theta \tau_p \frac{\delta_\theta^3 - \chi \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} [(1-r) + r\delta_\theta] \delta_\theta^2}{\left(\tau_\theta + \tau_y + 3\delta_\theta^2 \tau_p \right) - \chi \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} [2 + r(3\delta_\theta - 2)] \delta_\theta \tau_p + (1-r)\tau_s} + \delta_\theta^2 \\ &= \frac{\delta_\theta^2}{f_\delta} \left\{ -2\delta_\theta \tau_p \left(\delta_\theta - \chi \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} [(1-r) + r\delta_\theta] \right) + \left(\left(\tau_\theta + \tau_y + 3\delta_\theta^2 \tau_p \right) - \chi \frac{\tau_s [2 + r(3\delta_\theta - 2)]}{\tau_\theta + \tau_y + \tau_s} \delta_\theta \tau_p + (1-r)\tau_s \right) \right\} \\ &\propto \left(\tau_\theta + \tau_y + \delta_\theta^2 \tau_p \right) - \chi \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} r \delta_\theta^2 \tau_p + (1-r)\tau_s \\ &= \frac{1}{\delta_\theta} \left((1-r)\tau_s + \chi \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} (1-r)\delta_\theta^2 \tau_p \right) > 0 \end{aligned}$$

where we dropped $\frac{\delta_\theta^2}{f_\delta} > 0$ and, in the last step, we use $f(\delta_\theta) = 0$, which establishes the claim. $\square$

Proposition 11 (Proposition 2 w/o information acquisition). *In the model with exogenous τ_s , the weight on private information responds to parameter changes as follows:*

$$\frac{\partial \delta_\theta}{\partial \tau_s} \geq 0, \quad \frac{\partial \delta_\theta}{\partial \tau_y} \leq 0, \quad \frac{\partial \delta_\theta}{\partial r} \leq 0 \quad (52)$$

Proof. Since $f_\delta > 0$ (Lemma 5), for a generic parameter v we get $\frac{d\delta_\theta}{dv} = -\frac{f_v}{f_\delta} \propto -f_v$. And hence the comparative statics of δ_θ follow immediately, using (24), from

$$\begin{aligned} f_{\tau_y} &= \delta_\theta + \chi[(1-r) + r\delta_\theta] \delta_\theta^2 \tau_p \frac{\tau_s}{(\tau_\theta + \tau_y + \tau_s)^2} > 0 \\ f_{\tau_s} &= -\chi \frac{\tau_\theta + \tau_y}{(\tau_\theta + \tau_y + \tau_s)^2} [(1-r) + r\delta_\theta] \delta_\theta^2 \tau_p - (1-r)(1-\delta_\theta) < 0 \\ f_r &= \chi \frac{\tau_s(1-\delta_\theta)}{\tau_\theta + \tau_y + \tau_s} \delta_\theta^2 \tau_p + (1-\delta_\theta)\tau_s = \tau_s(1-\delta_\theta) \left[\frac{\chi \delta_\theta^2 \tau_p}{\tau_\theta + \tau_y + \tau_s} + 1 \right] > 0. \quad \square \end{aligned}$$

Proposition 12. *In the model with exogenous τ_s , we have*

$$\beta = 1 - \frac{\delta_\theta \tau_\theta}{(1-r)\tau_s}. \quad (53)$$

Furthermore,

$$\frac{d\beta}{d\chi} < 0, \quad \frac{d\beta}{d\tau_p} > 0, \quad \frac{d\beta}{d\tau_s} > 0, \quad \frac{d\delta_y}{d\tau_y} > 0, \quad \frac{d\delta_y}{d\tau_s} < 0, \quad \frac{d\delta_y}{d\tau_p} < 0.$$

Proof. The result for $\frac{d\beta}{d\chi}$, $\frac{d\beta}{d\tau_p}$, $\frac{d\delta_y}{d\tau_p}$ is immediate from the comparative statics of δ_θ . For $\frac{d\delta_y}{d\tau_y} \propto \frac{d\delta_\theta}{d\tau_y} \tau_y + \delta_\theta$, we get

$$\begin{aligned} \frac{d\delta_\theta}{d\tau_y} \tau_y + \delta_\theta &= -\frac{f_{\tau_y}}{f_\delta} \tau_y + \delta_\theta \propto -f_{\tau_y} \tau_y + f_\delta \delta_\theta \\ &= \frac{\delta_\theta}{(\tau_y + \tau_s + \tau_\theta)^2} \left[(3\delta_\theta^2 \tau_p + \tau_\theta + (1-r)\tau_s)(\tau_y + \tau_s + \tau_\theta)^2 - \chi \delta_\theta \tau_p \tau_s ((3-3r+4r\delta_\theta)\tau_y + (2-2r+3r\delta_\theta)(\tau_s + \tau_\theta)) \right] \\ &\propto \underbrace{(3\delta_\theta^2 \tau_p + \tau_\theta + (1-r)\tau_s)(\tau_y + \tau_s + \tau_\theta)^2 - 3\chi \delta_\theta \tau_p \tau_s (1-r+r\delta_\theta)(\tau_y + \tau_s + \tau_\theta)}_{=:A_1} + A_2 \end{aligned}$$

where, using (24),

$$A_1 = (\tau_y + \tau_s + \tau_\theta)^2 \left[\tau_\theta + (1-r)\tau_s + 3 \frac{(1-\chi)\delta_\theta^2 \tau_p (\tau_y + \tau_s + \tau_\theta)}{\tau_y + \tau_s + \tau_\theta + \chi \delta_\theta^2 \tau_p} \right] > 0.$$

It remains to show that $A_2 > 0$. We have

$$\begin{aligned} A_2 &\propto (1-r)(\tau_\theta + \tau_s) - \tau_y r \delta_\theta > (1-r)(\tau_\theta + \tau_s) - \tau_y r \delta_\theta^{\text{FC}} \\ &= (1-r)(\tau_\theta + \tau_s) - \tau_y r \frac{(1-r)\tau_s}{\tau_\theta + \tau_y + (1-r)\tau_s} = (1-r) \left(\tau_\theta + \tau_s \left(1 - r \frac{\tau_y}{\tau_\theta + \tau_y + (1-r)\tau_s} \right) \right) > 0. \end{aligned}$$

Note that the result for $\frac{d\delta_y}{d\tau_s}$ and $\frac{d\beta}{d\tau_s}$ follows when we establish $\frac{d}{d\tau_s} \frac{\delta_\theta}{\tau_s} < 0$:

$$\begin{aligned} \frac{d}{d\tau_s} \frac{\delta_\theta}{\tau_s} &= \frac{\frac{d\delta_\theta}{d\tau_s} \tau_s - \delta_\theta}{\tau_s^2} \propto \frac{d\delta_\theta}{d\tau_s} \tau_s - \delta_\theta \propto -f_{\tau_s} \tau_s + f_\delta \delta_\theta \\ &= (1-r)(1-\delta_\theta) \tau_s - \delta_\theta (\tau_y + \tau_\theta + (1-r) \tau_s) + 3 \frac{\chi(1-r) \tau_s \delta_\theta^2 \tau_p}{\tau_y + \tau_\theta + \tau_s} - \frac{\chi(1-r) \tau_s^2 \delta_\theta^2 \tau_p}{(\tau_y + \tau_\theta + \tau_s)^2} \\ &\quad - \delta_\theta^3 \tau_p \left(3 - 4 \frac{\chi r \tau_s}{\tau_y + \tau_\theta + \tau_s} + \frac{\chi r \tau_s^2}{(\tau_y + \tau_\theta + \tau_s)^2} \right) \end{aligned}$$

using f to replace $(1-r)(1-\delta_\theta) \tau_s$, we obtain a decomposition $B_1 + B_2$ where

$$\begin{aligned} B_1 &= \frac{\chi \delta_\theta^2 \tau_p \tau_s}{(\tau_y + \tau_\theta + \tau_s)^2} \left[\delta_\theta r (\tau_y + \tau_\theta) - (1-r) \tau_s \right] \\ B_2 &= -\delta_\theta (2\delta_\theta^2 \tau_p + (1-r) \tau_s) + 2 \frac{\chi \delta_\theta^2 [1-r + \delta_\theta r] \tau_p \tau_s}{\tau_y + \tau_\theta + \tau_s} \end{aligned}$$

Note that in B_1 the final term is negative if $r < 0$, otherwise, estimate $\delta_\theta < \delta_\theta^{\text{FC}}$ to arrive at

$$\begin{aligned} \delta_\theta r (\tau_y + \tau_\theta) - (1-r) \tau_s &< \frac{(1-r) \tau_s}{\tau_\theta + \tau_y + (1-r) \tau_s} r (\tau_y + \tau_\theta) - (1-r) \tau_s \\ &= -\frac{(1-r)^2 \tau_s (\tau_y + \tau_\theta + \tau_s)}{\tau_\theta + \tau_y + (1-r) \tau_s} < 0 \end{aligned}$$

and therefore $B_1 < 0$. Plugging (24) into B_2 , we arrive at

$$B_2 = -\delta_\theta \left((1-r) \tau_s + 2 \frac{(1-\chi) \delta_\theta^2 \tau_p (\tau_y + \tau_\theta + \tau_s)}{(\tau_y + \tau_\theta + \tau_s + \chi \delta_\theta^2 \tau_p)} \right) < 0,$$

whence we have established $\frac{d}{d\tau_s} \frac{\delta_\theta}{\tau_s} < 0$ and therefore $\frac{d\delta_y}{d\tau_s} < 0$ and $\frac{d\beta}{d\tau_s} > 0$. $\square$

C.3 Proofs for Section B (Shrewd Agent)

Proof of Proposition 6: This follows immediately from shrewd observation of the derivation of the matching coefficients and g equations. More directly, the welfare of an agent playing loadings α and private precision τ_s^R while the rest of (the average of) others play δ is derived in (25). By setting $\nabla_\alpha W(\alpha, \delta) = 0$ we find the best response coefficients as a function of others' loadings. We get

$$\begin{aligned} 0 &= (1 - \alpha_y - \alpha_p - \alpha_s + r(1 - \delta_\theta - \delta_y - \delta_p)) \tau_s - \alpha_s \tau_\theta \\ 0 &= (1 - \alpha_y - \alpha_p - \alpha_s + r(1 - \delta_\theta - \delta_y - \delta_p)) \tau_y - (\alpha_y - r \delta_y) \tau_\theta \end{aligned}$$

$$0 = (1 - \alpha_y - \alpha_p - \alpha_s + r(1 - \delta_\theta - \delta_y - \delta_p))\delta_\theta^2 \tau_p - (\alpha_p - r\delta_p)\tau_\theta$$

The solution to this linear system is the α^R in the proposition. Differentiating with respect to τ_s^R , we obtain the final condition. $\square$

Proof of Proposition 7: We first derive equation (44) in the text. Note that

$$\begin{aligned} \frac{(1 - (1 - \delta_\theta)r)^2}{(\tau_\theta + \tau_y + \tau_s^R + \delta_\theta^2 \tau_p)^2} - c &= 0 \\ (1 - (1 - \delta_\theta)r)^2 &= c [\tau_\theta + \tau_y + \tau_s^R + \delta_\theta^2 \tau_p]^2 \end{aligned}$$

using (24) on the LHS we get after rearranging

$$\left[\frac{(\tau_\theta + \tau_y + \tau_s)(\tau_\theta + \tau_y + \tau_s + \delta_\theta^2 \tau_p)}{\tau_\theta + \tau_y + \tau_s + \chi \delta_\theta^2 \tau_p} \right] = [\tau_\theta + \tau_y + \tau_s^R + \delta_\theta^2 \tau_p]$$

From there, $\tau_s \geq \tau_s^R$ is immediate, since

$$\frac{\tau_\theta + \tau_y + \tau_s + \delta_\theta^2 \tau_p}{\tau_\theta + \tau_y + \tau_s^R + \delta_\theta^2 \tau_p} = 1 + \frac{\chi \delta_\theta^2 \tau_p}{\tau_\theta + \tau_y + \tau_s} \geq 1$$

For the fully cursed equilibrium, the above implies that $\tau_\theta + \tau_y + \tau_s = \tau_\theta + \tau_y + \tau_s^R + \delta_\theta^2 \tau_p$ and hence we have

$$\tau_s^R = \tau_s - \delta_\theta^2 \tau_p = \frac{1}{\sqrt{c}} \frac{1 - r - \sqrt{c}(\tau_\theta + \tau_y)}{1 - r} - \tau_p \left(\frac{1 - r - \sqrt{c}(\tau_\theta + \tau_y)}{1 - r} \right)^2$$

Solving for $\tau_s^R \geq 0$, we get positive information acquisition iff

$$\tau_\theta + \tau_y \in \left(\frac{(1 - r) \left(1 - \frac{1}{\tau_p \sqrt{c}} \right)}{\sqrt{c}}, \frac{1 - r}{\sqrt{c}} \right) \quad (54)$$

Now, for the Proposition, we have established $\tau_s^R \leq \tau_s$. For the limit result, note that

$$\tau_s^R = \frac{(\tau_\theta + \tau_y + \tau_s)(\tau_\theta + \tau_y + \tau_s + \delta_\theta^2 \tau_p)}{\tau_\theta + \tau_y + \tau_s + \chi \delta_\theta^2 \tau_p} - (\tau_\theta + \tau_y + \delta_\theta^2 \tau_p) = \frac{(\tau_\theta + \tau_y + \tau_s)\tau_s - \chi \delta_\theta^2 \tau_p (\tau_\theta + \tau_y + \delta_\theta^2 \tau_p)}{\tau_\theta + \tau_y + \tau_s + \chi \delta_\theta^2 \tau_p}$$

and that $\tau_s \rightarrow \infty$ as $c \rightarrow 0$. From there, we have

$$\tau_s^R = \frac{(\tau_\theta + \tau_y + \tau_s)\tau_s - \chi\delta_\theta^2\tau_p(\tau_\theta + \tau_y + \delta_\theta^2\tau_p)}{\tau_\theta + \tau_y + \tau_s + \chi\delta_\theta^2\tau_p} \rightarrow \frac{\tau_s^2}{\tau_s} \rightarrow \infty$$

We have $\tau_s^R = 0$ for $\tau_\theta + \tau_y \leq \frac{(1-r)\left(1-\frac{1}{\tau_p\sqrt{c}}\right)}{\sqrt{c}}$ in the fully cursed case, where $\tau_s > 0$.

To see nonmonotonicity in $\tau_\theta + \tau_y$ in the general case, note that as we approach the limit (15), we have

$$\frac{d\tau_s^R}{d\tau_y}\bigg|_{\sqrt{c}=\frac{1-r}{\tau_\theta+\tau_y}} = \frac{\tau_\theta + \tau_y}{\sqrt{c}(2\tau_\theta + \tau_y)} \frac{d\delta_\theta}{d\tau_y} \propto \frac{d\delta_\theta}{d\tau_y} < 0$$

and hence, local to this value, we always get a positive τ_s^R . However, for $\tau_\theta + \tau_y \leq \chi\frac{1-r}{\sqrt{c}}$ interior, we get that for $\tau_p \rightarrow \infty$

$$\tau_s^R = \frac{(\tau_\theta + \tau_y + \tau_s)\tau_s - \chi\delta_\theta^2\tau_p(\tau_\theta + \tau_y + \delta_\theta^2\tau_p)}{\tau_\theta + \tau_y + \tau_s + \chi\delta_\theta^2\tau_p} \rightarrow -\delta_\theta^2\tau_p < 0$$

which establishes the result. Interior nonmonotonicity follows by continuity.

To see nonmonotonicity in c , note that

1. $\frac{d\tau_s^R}{dc} < 0$ in the limit (15) ($\sqrt{c} = \frac{1-r}{\tau_\theta+\tau_y}$). To see this, take the representation above, plug 13, take the derivative, set $\sqrt{c} = \frac{1-r}{\tau_\theta+\tau_y}$, $\delta_\theta = 0$, then we get

$$\frac{d\tau_s^R}{dc}\bigg|_{\sqrt{c}=\frac{1-r}{\tau_\theta+\tau_y}} \rightarrow \frac{2(1-r)^4 \frac{\partial\delta_\theta}{\partial c}}{(\tau_\theta + \tau_y)^2} \propto \frac{\partial\delta_\theta}{\partial c} < 0$$

2. $\tau_s^R = 0$ at the limit (15), as $0 \leq \tau_s^R \leq \tau_s = 0$. Combining this with (1.), we obtain that $\tau_s^R > 0$ local to this upper bound on costs.
3. $\tau_s^R \rightarrow \infty$ as $c \rightarrow 0$, as shown above.
4. For any c satisfying the existence of an interior equilibrium in the $\tau_p \rightarrow \infty$ limit (i.e. $\sqrt{c} \leq \chi\frac{1-r}{\tau_\theta+\tau_y}$, see footnote 24), there exists a τ_p sufficiently large such that $(\tau_s^R)^{\text{FOC}} < 0$. To see this, pick an interior c . Then, because $\delta_\theta^\infty > 0$ and $\tau_p \rightarrow \infty$

$$(\tau_s^R)^{\text{FOC}} \rightarrow -\delta_\theta^2\tau_p < 0$$

This establishes nonmonotonicity, as τ_s^R is large for small c , zero for intermediate c , but nonzero local to $\sqrt{c} = \frac{1-r}{\tau_\theta+\tau_y}$.

To analyse the derivative in χ , we compute

$$\frac{d\tau_s^R}{d\chi} \propto r - 2\sqrt{c}\delta_\theta\tau_p$$

It is apparent that τ_s^R is decreasing for $r \leq 0$ and that it is increasing as $r \rightarrow 1 - \sqrt{c}(\tau_\theta + \tau_y)$ when this is positive, as then $\delta_\theta \rightarrow 0$. To see that we can have a hump shape, note that δ_θ is increasing in χ and hence

$$\frac{d^2\tau_s^R}{d\chi d\chi} \propto -\frac{d\delta_\theta}{d\chi} \leq 0$$

around $\frac{d\tau_s^R}{d\chi} = 0$, which establishes a hump-shape (but, importantly, not necessarily concavity!). Clearly, all these comparative statics only apply for interior solutions, otherwise $\tau_s^R \equiv 0$ locally. $\square$

Proof of Proposition 8: For χ , we plug the general α^R, δ into the welfare expression, take derivative w.r.t. χ , set $\chi = 1$, $\delta_\theta = \delta_\theta^{FC}$ and we get

$$\begin{aligned} \frac{dW_\chi^R}{d\chi} &= -\frac{2\sqrt{c}(1-r)\tau_p(1-r-\sqrt{c}(\tau_\theta+\tau_y))^2[r(\tau_\theta+\tau_y+\tau_s^R+\tau_p)-\tau_p(1-\sqrt{c}(\tau_\theta+\tau_y))](1-r(1+\sqrt{c}(\tau_\theta+\tau_y)))^2}{[\cdot]^2\left[1+\sqrt{c}\tau_p(1-\sqrt{c}(\tau_\theta+\tau_y))^2+r^2(1+\sqrt{c}(\tau_\theta+\tau_y+\tau_p))-r(2-2c\tau_p(\tau_\theta+\tau_y)+\sqrt{c}(\tau_\theta+\tau_y+2\tau_p))\right]} \\ &\propto -\frac{2\sqrt{c}\tau_p\delta_\theta^2(1-r)^3[r(\tau_\theta+\tau_y+\tau_s^R+\tau_p\delta_\theta)-\tau_p\delta_\theta]}{1+\sqrt{c}\tau_p(1-\sqrt{c}(\tau_\theta+\tau_y))^2+r^2(1+\sqrt{c}(\tau_\theta+\tau_y+\tau_p))-r(2-2c\tau_p(\tau_\theta+\tau_y)+\sqrt{c}(\tau_\theta+\tau_y+2\tau_p))} \end{aligned} \quad (55)$$

Consider first the case of $r > 0$, which implies that $1 - \sqrt{c}(\tau_\theta + \tau_y) > 0$. As $r \rightarrow 1 - \sqrt{c}(\tau_\theta + \tau_y)$, we have for the above expression

$$\rightarrow -2\tau_p(\tau_\theta + \tau_y)\sqrt{c}(1 - \sqrt{c}(\tau_\theta + \tau_y))\delta_\theta^2 \leq 0$$

This converges to zero from below since $\delta_\theta \rightarrow 0$ as $r \rightarrow 1 - \sqrt{c}(\tau_\theta + \tau_y)$. Therefore, for large χ , the shrewd agent prefers a less cursed environment if r is big.

Consider now $r \leq 0$. Note that the numerator in (55) is always negative. Furthermore, note that the denominator $v(r)$ is positive for $r = 0$. We will show that it is positive for all $r \leq 0$ and hence the expression is positive for all $r \leq 0$. First, note that v is a convex quadratic function in r . Minimizing, we find that the minimizer and minimum, resp., are given by

$$\begin{aligned} r^* &\propto 2 - 2c\tau_p(\tau_\theta + \tau_y) + \sqrt{c}(\tau_\theta + \tau_y + 2\tau_p) \\ v^* &\propto -1 + 4c\tau_p(\tau_\theta + \tau_y) \end{aligned}$$

If $v^* > 0$, we are done. If it is negative, it is easy to see that r^* must be positive. But then,

the fact that $v(0) > 0$ implies that $v(r) > 0$ for all $r \leq 0$. $\square$

Proof of Proposition 9: To obtain the welfare of the shrewd agent in the fully cursed equilibrium, we simply plug action rule (40)-(43) and the equilibrium δ s into the welfare equation to obtain, for the unconstrained case

$$W_1^{R, \tau_s^R > 0} = -2\sqrt{c} + \frac{-2c^{3/2}(1-r)\tau_p(\tau_\theta + \tau_y) + c^2\tau_p(\tau_\theta + \tau_y)^2 + c(1-r)(\tau_\theta + \tau_y + (1-r)\tau_p)}{(1-r)^2}$$

as well as for the constrained case, where we leave the expression in general form both for compactness and ease of analysis

$$W_1^{R, \tau_s^R = 0} = -\frac{(1-r)r\delta_y^2}{\tau_y} - \frac{(1-r)r(1-\delta_\theta - \delta_y)^2}{\tau_\theta} - \frac{(1-(1-\delta_\theta)r)^2}{\tau_\theta + \tau_y + \delta_\theta^2\tau_p}$$

For transparency, we obtain by direct computation for an interior solution $\frac{\partial W_1^{R, \tau_s^R > 0}}{\partial \tau_p} = c \frac{(1-r-\sqrt{c}(\tau_\theta + \tau_y))^2}{(1-r)^2} = c\delta_\theta^2 > 0$, and through an envelope argument from the general expres-

sion, we get form corner solutions $\frac{\partial W_1^{R, \tau_s^R = 0}}{\partial \tau_p} = \frac{\delta_\theta^2(1-(1-\delta_\theta)r)^2}{(\tau_\theta + \tau_y + \delta_\theta^2\tau_p)^2} > 0$.

For τ_θ, τ_y : Consider the derivative of $W_1^{R, \tau_s^R > 0}$ and let $\tau_\theta + \tau_y \rightarrow \frac{(1-r)(1-\frac{1}{\tau_p\sqrt{c}})}{\sqrt{c}}$. Then, we get

$$\frac{\partial W_1^{R, \tau_s^R > 0}}{\partial \tau_y} \rightarrow c \frac{1-r-(1-r)2\sqrt{c}\tau_p + 2c\tau_p(\tau_\theta + \tau_y)}{(1-r)^2} = \frac{c}{1-r} (1 - 2\sqrt{c}\tau_p\delta_\theta^{FC}) = -\frac{c}{1-r} < 0$$

Note that this result also holds for $W_1^{R, \tau_s^R = 0}$, as the value function is $\mathcal{C}^1$. Taking instead the limit as $\tau_\theta + \tau_y \rightarrow \frac{(1-r)}{\sqrt{c}}$ (where we always are at an interior solution), we have

$$\frac{\partial W_1^{R, \tau_s^R > 0}}{\partial \tau_y} \rightarrow \frac{c}{1-r} (1 - 2\sqrt{c}\tau_p\delta_\theta^{FC}) = \frac{c}{1-r}$$

For c , let us first demonstrate a setting where an increase in costs is beneficial for the shrewd agent. Consider constrained welfare, let $r \rightarrow -\infty$ and hence $\delta_\theta = 1 - \frac{\sqrt{c}(\tau_\theta + \tau_y)}{(1-r)} \rightarrow 1$, $\delta_y = \frac{\sqrt{c}}{1-r}\tau_y \rightarrow 0$. Then

$$W_1^{R, \tau_s^R = 0} \rightarrow -\frac{rc\tau_y}{(1-r)} - \frac{r\tau_\theta c}{(1-r)} - \frac{\left(1 - \frac{\sqrt{c}(\tau_\theta + \tau_y)}{(1-r)}r\right)^2}{\tau_\theta + \tau_y + \tau_p} \rightarrow c\tau_\theta + \tau_y c - \frac{\left(1 + \sqrt{c}(\tau_\theta + \tau_y)\right)^2}{\tau_\theta + \tau_y + \tau_p}$$

and

$$\begin{aligned}\frac{\partial}{\partial c} \left(c\tau_\theta + \tau_y c - \frac{(1 + \sqrt{c}(\tau_\theta + \tau_y))^2}{\tau_\theta + \tau_y + \tau_p} \right) &= \tau_\theta + \tau_y - \frac{(\tau_\theta + \tau_y)(1 + \sqrt{c}(\tau_\theta + \tau_y))}{\sqrt{c}(\tau_\theta + \tau_y + \tau_p)} \\ &= (\tau_\theta + \tau_y) \left(\frac{\tau_p - \frac{1}{\sqrt{c}}}{\tau_\theta + \tau_y + \tau_p} \right) > 0\end{aligned}$$

since we consider the case $\tau_s^R = 0$, and therefore $\tau_p > \frac{1}{\sqrt{c}}$.

Generally, we can also have $\frac{\partial}{\partial c} W_1^R < 0$. To see this, consider the case of $r = 0$. Then

$$W^R|_{r=0} = \max_{\tau_s^R \geq 0} -\frac{1}{\tau_\theta + \tau_y + \tau_s^R + \delta_\theta^2 \tau_p} - c\tau_s^R$$

where we may or may not have a corner solution. In either case, welfare is decreasing in c because δ_θ is decreasing in c and an envelope argument.

At an interior solution (which occurs for $\tau_p < \frac{1}{\delta_\theta \sqrt{c}}$), we get

$$\begin{aligned}\frac{\partial}{\partial c} W_1^{R, \tau_s^R > 0} &= -\frac{1}{\sqrt{c}} + \frac{-3\sqrt{c}(1-r)\tau_p(\tau_\theta + \tau_y) + 2c\tau_p(\tau_\theta + \tau_y)^2 + (1-r)(\tau_\theta + \tau_y + (1-r)\tau_p)}{(1-r)^2} \\ &= \frac{(\tau_\theta + \tau_y)}{(1-r)} - \frac{1}{\sqrt{c}} + \delta_\theta \tau_p \left(\delta_\theta - \frac{\sqrt{c}(\tau_\theta + \tau_y)}{(1-r)} \right)\end{aligned}$$

which is always negative by condition (15) if $\delta_\theta \leq \frac{\sqrt{c}(\tau_\theta + \tau_y)}{(1-r)}$. If this is violated, we have

$$\begin{aligned}\frac{(\tau_\theta + \tau_y)}{(1-r)} - \frac{1}{\sqrt{c}} + \delta_\theta \tau_p \left(\delta_\theta - \frac{\sqrt{c}(\tau_\theta + \tau_y)}{(1-r)} \right) &\leq \frac{(\tau_\theta + \tau_y)}{(1-r)} - \frac{1}{\sqrt{c}} + \frac{1}{\sqrt{c}} \left(\delta_\theta - \frac{\sqrt{c}(\tau_\theta + \tau_y)}{(1-r)} \right) \\ &= -\frac{\delta_\theta}{\sqrt{c}} + \frac{1}{\sqrt{c}} \left(\delta_\theta - \frac{\sqrt{c}(\tau_\theta + \tau_y)}{(1-r)} \right) = \frac{1}{\sqrt{c}} \left(-\frac{\sqrt{c}(\tau_\theta + \tau_y)}{(1-r)} \right) < 0\end{aligned}$$

concluding the proof. $\square$

C.4 Proofs of the Lemmata in Appendix A (Proof Appendix)

Lemma 6. *There is no equilibrium with information acquisition and $\delta_\theta < 0$.*

Proof of Lemma 6: Towards a contradiction, let $\delta_\theta < 0$. Then, $\tau_s = -\frac{\delta_\theta}{\sqrt{c}}$ and f reads

$$(\tau_\theta + \tau_y + \delta_\theta^2 \tau_p) + \chi \frac{1}{\sqrt{c}(\tau_\theta + \tau_y) - \delta_\theta} [(1-r) + r\delta_\theta] \delta_\theta^2 \tau_p + \frac{1}{\sqrt{c}} (1-r)(1-\delta_\theta) = 0$$

Clearly, for $\delta_\theta = 0$, the expression is strictly positive. Furthermore, we have $\frac{d}{d\delta_\theta} f\left(\delta_\theta, -\frac{\delta_\theta}{\sqrt{c}}\right) < 0$, as

$$\begin{aligned}
& 2\delta_\theta \tau_p + \chi \frac{[(1-r) + r\delta_\theta] \delta_\theta^2 \tau_p}{(\sqrt{c}(\tau_\theta + \tau_y) - \delta_\theta)^2} + \chi \frac{[(1-r) + 3r\delta_\theta^2] \tau_p}{\sqrt{c}(\tau_\theta + \tau_y) - \delta_\theta} - \frac{1}{\sqrt{c}}(1-r) \\
&= \frac{1}{(\sqrt{c}(\tau_\theta + \tau_y) - \delta_\theta)^2} \left[2\delta_\theta \tau_p (\sqrt{c}(\tau_\theta + \tau_y) - \delta_\theta)^2 + \chi [(1-r) + r\delta_\theta] \delta_\theta^2 \tau_p \right. \\
&\quad \left. + \chi (\sqrt{c}(\tau_\theta + \tau_y) - \delta_\theta) \left[2(1-r)\delta_\theta + 3r\delta_\theta^2 \right] \tau_p - \frac{1}{\sqrt{c}}(1-r)(\sqrt{c}(\tau_\theta + \tau_y) - \delta_\theta)^2 \right] \\
&\propto 2\delta_\theta^3 \tau_p - 4\delta_\theta \tau_p \sqrt{c}(\tau_\theta + \tau_y) + 2\delta_\theta \tau_p c(\tau_\theta + \tau_y)^2 + \chi [(1-r) + r\delta_\theta] \delta_\theta^2 \tau_p \\
&\quad + \chi (\sqrt{c}(\tau_\theta + \tau_y) - \delta_\theta) \left[2(1-r)\delta_\theta + 3r\delta_\theta^2 \right] \tau_p - \frac{1}{\sqrt{c}}(1-r)(\sqrt{c}(\tau_\theta + \tau_y) - \delta_\theta)^2 \\
&= 2\delta_\theta^3 \tau_p (1 - \chi r) - \delta_\theta^2 \left[\frac{1}{\sqrt{c}}(1-r) + (4\tau_p - 3\chi r \tau_p) \sqrt{c}(\tau_\theta + \tau_y) + \chi(1-r)\tau_p \right] \\
&\quad + \delta_\theta 2c(\tau_\theta + \tau_y) \left[(1-r)(1 + \chi \tau_p \sqrt{c}) \tau_p + \tau_p c(\tau_\theta + \tau_y) \right] - \sqrt{c}(1-r)(\tau_\theta + \tau_y)^2 < 0
\end{aligned}$$

Hence, we cannot have an solution, as $f > 0$ for all $\delta_\theta < 0$ and there is no such equilibrium. $\square$

Proof of Lemma 1: Suppose (15) holds. Then, using (24) and (13), we have

$$\begin{aligned}
\frac{d}{d\delta_\theta} \tilde{f}(\delta_\theta) &= \frac{(1-r)(\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y)) + \sqrt{c}\delta_\theta^2 \tau_p \chi \left[\frac{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y + \delta_\theta^2 \tau_p)}{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y + \chi \delta_\theta^2 \tau_p)} - r \right]}{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y)} + 2\sqrt{c}\delta_\theta \tau_p \left((1-\chi) \frac{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y)}{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y + \chi \delta_\theta^2 \tau_p)} \right) \\
&\geq \frac{(1-r)(\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y)) + \sqrt{c}\delta_\theta^2 \tau_p \chi \left[\frac{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y + \delta_\theta^2 \tau_p)}{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y + \chi \delta_\theta^2 \tau_p)} - r \right]}{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y)} \geq \frac{\sqrt{c}\delta_\theta^2 \tau_p \chi [1-r]}{\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y)} > 0
\end{aligned}$$

since the first term in the numerator is positive and the fraction in square brackets is greater than 1. $\square$

Proof of Lemma 2: Note that $g_\delta = 2\delta_\theta > 0$, $g_{\tau_s} = -2c\tau_s < 0$ and

$$\begin{aligned}
f_\delta &= (\tau_\theta + \tau_y + 3\delta_\theta^2 \tau_p) - \chi \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} [2 + r(3\delta_\theta - 2)] \delta_\theta \tau_p + (1-r)\tau_s \\
f_{\tau_s} &= -\chi \frac{\tau_\theta + \tau_y}{(\tau_\theta + \tau_y + \tau_s)^2} [1 + r(\delta_\theta - 1)] \delta_\theta^2 \tau_p - (1-r)(1 - \delta_\theta).
\end{aligned}$$

By direct computation

$$g_\delta f_{\tau_s} - g_{\tau_s} f_\delta = 2\delta_\theta \left(-\chi \frac{\tau_\theta + \tau_y}{(\tau_\theta + \tau_y + \tau_s)^2} [1 + r(\delta_\theta - 1)] \delta_\theta^2 \tau_p - (1-r)(1 - \delta_\theta) \right)$$

$$\begin{aligned}
& -(-2c\tau_s) \left((\tau_\theta + \tau_y + 3\delta_\theta^2 \tau_p) - \chi \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} [2 + r(3\delta_\theta - 2)] \delta_\theta \tau_p + (1-r)\tau_s \right) \\
& = 2\delta_\theta \left(-\chi \frac{\tau_\theta + \tau_y}{(\tau_\theta + \tau_y + \tau_s)^2} [1 + r(\delta_\theta - 1)] \delta_\theta^2 \tau_p - [1 + r(\delta_\theta - 1)] + \delta_\theta \right) \\
& \quad + 2 \frac{\delta_\theta}{\tau_s} \left((\tau_\theta + \tau_y + 3\delta_\theta^2 \tau_p) \delta_\theta - 2\chi \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} [1 + r(\delta_\theta - 1)] \delta_\theta^2 \tau_p + (1-r)\tau_s \delta_\theta - \chi \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} r \delta_\theta \delta_\theta^2 \tau_p \right) \\
& \stackrel{(f_\delta > 0)}{\geq} 2 \frac{\delta_\theta}{\tau_s} \left(-\chi \frac{\tau_\theta + \tau_y}{\tau_\theta + \tau_y + \tau_s} \frac{(\tau_\theta + \tau_y + \tau_s + \delta_\theta^2 \tau_p) \delta_\theta}{\tau_\theta + \tau_y + \tau_s + \chi \delta_\theta^2 \tau_p} \delta_\theta^2 \tau_p - \frac{\delta_\theta (\tau_\theta + \tau_y + \tau_s) (\tau_\theta + \tau_y + \tau_s + \delta_\theta^2 \tau_p)}{\tau_\theta + \tau_y + \tau_s + \chi \delta_\theta^2 \tau_p} + \delta_\theta \tau_s \right) \\
& \quad + 2 \frac{\delta_\theta}{\tau_s} \left(\left(\tau_\theta + \tau_y + 3 \left(1 - \chi \frac{\tau_\theta + \tau_y + \tau_s + \delta_\theta^2 \tau_p}{\tau_\theta + \tau_y + \tau_s + \chi \delta_\theta^2 \tau_p} \right) \delta_\theta^2 \tau_p \right) \delta_\theta + (1-r)\tau_s \delta_\theta + \chi \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} [1-r] \delta_\theta^2 \tau_p \right) \\
& = 2\delta_\theta^2 \left\{ -\frac{\tau_y}{\tau_s} - \frac{\tau_\theta}{\tau_s} + \delta_\theta^2 \tau_p \left(-\frac{1}{\tau_s} + \frac{1}{\tau_\theta + \tau_y + \tau_s} - \frac{1-\chi}{\tau_\theta + \tau_y + \tau_s + \chi \delta_\theta^2 \tau_p} \right) \right\} \\
& \quad + 2 \frac{\delta_\theta}{\tau_s} \left(\left(\tau_\theta + \tau_y + 3 \left(1 - \chi \frac{\tau_\theta + \tau_y + \tau_s + \delta_\theta^2 \tau_p}{\tau_\theta + \tau_y + \tau_s + \chi \delta_\theta^2 \tau_p} \right) \delta_\theta^2 \tau_p \right) \delta_\theta + (1-r)\tau_s \delta_\theta + \chi \frac{\tau_s}{\tau_\theta + \tau_y + \tau_s} [1-r] \delta_\theta^2 \tau_p \right) \\
& \stackrel{(r < 1 \& (13))}{\geq} 2\delta_\theta^3 \tau_p (1-\chi) \sqrt{c} \left\{ \frac{2\delta_\theta^2 + \sqrt{c}\chi\delta_\theta^3 \tau_p + 4\sqrt{c}\delta_\theta (\tau_\theta + \tau_y) + 2c(\tau_\theta + \tau_y)^2}{[\delta_\theta + \sqrt{c}(\tau_\theta + \tau_y)][\delta_\theta + \sqrt{c}\chi\delta_\theta^2 \tau_p + \sqrt{c}(\tau_\theta + \tau_y)]} \right\} \geq 0
\end{aligned}$$

The last equality follows from lengthy but straightforward calculation. The inequality follows since the expression is decreasing in r and we hence set $r = 1$ as a worst case, obtaining our result. $\square$

Proof of Lemma 3: We have

$$\begin{aligned}
f_\delta^\star &= \frac{[2(\tau_\theta + \tau_y + \delta_\theta^2 \tau_p) \delta_\theta \tau_p] (\tau_\theta + \tau_y + \tau_p \delta_\theta) - \tau_p (\tau_\theta + \tau_y + \delta_\theta^2 \tau_p)^2}{(\tau_\theta + \tau_y + \tau_p \delta_\theta)^2} + (1-r) \frac{1}{\sqrt{c}} \\
&= \frac{\tau_p (\tau_\theta + \tau_y + \delta_\theta^2 \tau_p)}{(\tau_\theta + \tau_y + \tau_p \delta_\theta)^2} \left[\frac{(1-r)}{\sqrt{c}} \frac{(\tau_\theta + \tau_y + \tau_p \delta_\theta)^2}{\tau_p (\tau_\theta + \tau_y + \delta_\theta^2 \tau_p)} - [2\delta_\theta - 1] (\tau_\theta + \tau_y) + \delta_\theta^2 \tau_p \right] \\
&\geq \frac{\tau_p (\tau_\theta + \tau_y + \delta_\theta^2 \tau_p)}{(\tau_\theta + \tau_y + \tau_p \delta_\theta)^2} \left[\frac{(1-r)}{\sqrt{c}} \left(\frac{(\tau_\theta + \tau_y + \tau_p \delta_\theta)^2}{\tau_p (\tau_\theta + \tau_y + \delta_\theta^2 \tau_p)} - [2\delta_\theta - 1] \right) + \delta_\theta^2 \tau_p \right]
\end{aligned}$$

using that $\frac{1-r}{\tau_\theta + \tau_y} > \sqrt{c}$. Then, from

$$\begin{aligned}
\frac{(\tau_\theta + \tau_y + \tau_p \delta_\theta)^2}{\tau_p (\tau_\theta + \tau_y + \delta_\theta^2 \tau_p)} - [2\delta_\theta - 1] &\propto \tau_p (\tau_\theta + \tau_y + \delta_\theta^2 \tau_p) + (\tau_\theta + \tau_y + \tau_p \delta_\theta)^2 - 2\delta_\theta \tau_p (\tau_\theta + \tau_y + \delta_\theta^2 \tau_p) \\
&= \tau_p (\tau_\theta + \tau_y) + (\tau_\theta + \tau_y)^2 + (2 - 2\delta_\theta) \tau_p \delta_\theta^2 \tau_p > 0
\end{aligned}$$

we obtain the result. $\square$

Proof of Lemma 4: Taking the $\tau_p \rightarrow \infty$ limit in (14) we get the limit reliance on private information is²⁴

$$\delta_\theta^\infty = \frac{\chi(1-r) - (\tau_\theta + \tau_y)\sqrt{c}}{(1-r\chi)}, \quad (56)$$

which we can plug in (21) to obtain that welfare in the limit equilibrium is given by

$$W^\infty = -\frac{2\sqrt{c}\chi(1-r)^2 - c(1-2r+r\chi)(\tau_\theta + \tau_y)}{(1-r)(1-r\chi)}, \quad (57)$$

which we can directly differentiate. To see the comparative static, note that the coefficient of $\tau_\theta + \tau_y$ is $1-2r+r\chi$. Consider the case where $r < 0$, then this expression is negative only for $\chi > 2$, so this case is irrelevant. Instead, with $r > 0$, we get that the impact of τ_θ, τ_y is negative iff $\chi \leq 2 - \frac{1}{r}$.

The derivative w.r.t. costs is

$$\frac{\partial W^\infty}{\partial c} \propto -\frac{\chi}{\sqrt{c}}(1-r)^2 + (1-2r+r\chi)(\tau_\theta + \tau_y)$$

and

$$\begin{aligned} -\frac{\chi}{\sqrt{c}}(1-r)^2 + (1-2r+r\chi)(\tau_\theta + \tau_y) &\geq 0 \\ \chi[(1-r)^2 - r\sqrt{c}(\tau_\theta + \tau_y)] &\leq \sqrt{c}(1-2r)(\tau_\theta + \tau_y) \end{aligned}$$

Hence, there are two cases we need to consider. First, if $(1-r)^2 - r\sqrt{c}(\tau_\theta + \tau_y) < 0$: Note that this can only be the case if $r > \frac{1}{2}$, since otherwise by (15)

$$(1-r)^2 - r\sqrt{c}(\tau_\theta + \tau_y) \geq (1-r)^2 - r(1-r) = (1-2r)(1-r) > 0.$$

Then, we obtain a lower bound for χ :

$$\chi \geq \sqrt{c}(1-2r) \frac{\tau_\theta + \tau_y}{(1-r)^2 - r\sqrt{c}(\tau_\theta + \tau_y)}$$

²⁴Because the optimality condition (13) implies that τ_s^∞ solves $\tau_s\sqrt{c} = \frac{\chi\tau_s(1-r)}{\tau_\theta + \tau_y + \tau_s(1-r\chi)}$, an interior limit equilibrium exists if (and only if) $\chi > \sqrt{c} \frac{\tau_\theta + \tau_y}{(1-r)}$, a condition that we assume satisfied. Notice that this condition does not contradict the sufficiency of condition (15) which holds for any interior τ_p . The order of limits is relevant, as for every fixed τ_p , the influence of transparency vanishes as δ_θ and τ_s go to zero. All details about the derivations of the transparent limit are available upon request.

but this bound rules out that it is increasing in costs, since

$$\sqrt{c}(1-2r) \frac{\tau_\theta + \tau_y}{(1-r)^2 - r\sqrt{c}(\tau_\theta + \tau_y)} \geq 1 \iff \sqrt{c}(\tau_\theta + \tau_y) \leq (1-r)$$

which is guaranteed by (15).

Second, if $(1-r)^2 - r\sqrt{c}(\tau_\theta + \tau_y) > 0$: we get an upper bound

$$\chi \leq \sqrt{c}(1-2r) \frac{\tau_\theta + \tau_y}{(1-r)^2 - r\sqrt{c}(\tau_\theta + \tau_y)}.$$

The resulting interval is nontrivial only if $r < 0$, as by (15)

$$\begin{aligned} \sqrt{c} \frac{\tau_\theta + \tau_y}{1-r} &< \sqrt{c}(1-2r) \frac{\tau_\theta + \tau_y}{(1-r)^2 - r\sqrt{c}(\tau_\theta + \tau_y)} \\ r \left[\frac{\sqrt{c}(\tau_\theta + \tau_y)}{(1-r)} \right] &> r \iff r < 0. \end{aligned}$$

□